

Ultrasonic Imaging Systems

23.1 DIAGNOSTIC ULTRASOUND

Ultrasound has become increasingly important in medicine and has taken its place along with X-ray and nuclear medicine as a diagnostic tool. Its main attraction as an imaging modality lies in its non-invasive character and ability to distinguish interfaces between soft tissues. In contrast, X-rays only respond to atomic weight differences and often require the injection of a denser contrast medium for visualization of non-bony tissues. Similarly, nuclear medicine techniques measure the selective uptake of radioactive isotopes in specific organs to produce information concerning organ function. Radioactive isotopes and X-rays are, thus, clearly invasive. Ultrasound is not only non-invasive, externally applied and non-traumatic but also apparently safe at the acoustical intensities and duty cycles presently used in diagnostic equipment.

Diagnostic ultrasound is applied for obtaining images of almost the entire range of internal organs in the abdomen. These include the kidney, liver, spleen, pancreas, bladder, major blood vessels and of course, the foetus during pregnancy. It has also been usefully employed to present pictures of the thyroid gland, the eyes, the breasts and a variety of other superficial structures. In a number of medically meaningful cases, ultrasonic diagnostics has made possible the detection of cysts, tumours or cancer in these organs. This is possible in structures where other diagnostic methods by themselves were found to be either inapplicable, insufficient or unacceptably hazardous. Ultrasonic studies which do not involve image formation have also been extensively developed to allow the dynamics of blood flow in the cardiovascular system to be investigated with a precision not previously possible. The main limitation of ultrasound, however, is that it is almost completely reflected at boundaries with gas and is a serious restriction in investigation of and through gas-containing structures.

23.2 PHYSICS OF ULTRASONIC WAVES

Ultrasonic waves are sound waves associated with frequencies above the audible range and generally extend upward from 20 kHz. These waves exhibit the same physical properties as the audible sound waves but they are particularly preferred in situations favoured by one or more of the following reasons:

- Ultrasonic waves can be easily focussed, i.e., they are directional and beams can be obtained with very little spreading.

- They are inaudible and are suitable for applications where it is not advantageous to employ audible frequencies.
- By using high frequency ultrasonic waves which are associated with shorter wavelengths, it is possible to investigate the properties of very small structures. It is particularly true in the detection of defects where the wavelengths utilized should be of the same order as the dimensions of the defect.
- Information obtained by ultrasound, particularly in dynamic studies, cannot be acquired by any other more convenient technique.

Transmission of ultrasonic wave motion can take place in different modes. The wave motion may be longitudinal, transverse or shear. However, for medical ultrasonic diagnostic applications, the longitudinal mode of wave propagation is normally used as these waves can be propagated in all types of media, viz. solids, liquids and gases. In longitudinal waves, the particles of the medium oscillate to and fro in the direction of propagation of the wave resulting in alternate regions of compressions and rarefactions.

23.2.1 Characteristic Impedance

Characteristic impedance or the specific acoustic impedance of a medium is defined as the product of the density of the medium with the velocity of sound in the same medium.

$$z = \rho V$$

where z = specific acoustic impedance

ρ = density of the medium

V = velocity of sound in the medium.

The characteristic impedance determines the degree of reflection and refraction at the interface between two media. The percent of the incident wave energy which is reflected is given by:

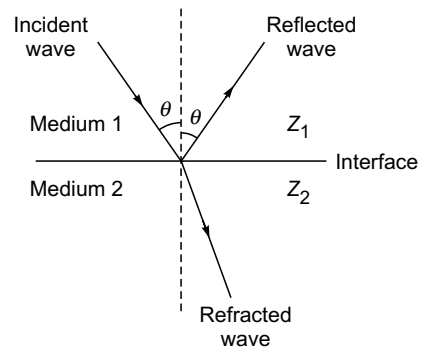
$$\left[\frac{z_1 - z_2}{z_1 + z_2} \right]^2 \times 100\%$$

where z_1 = acoustic impedance of medium 1

z_2 = acoustic impedance of medium 2

provided the ultrasonic beam strikes the interface at a right angle. Beams incident on a boundary at an angle (Fig. 23.1) other than zero to the normal are reflected so that the angle of incidence and the angle of reflection are equal.

The approximate value of acoustic impedance for most of the biological materials or organs is the same. It is about $1.6 \times 10^5 \text{ g/cm}^2\text{s}$. The greater the difference in acoustic impedance, the greater the amount of reflected energy. For example, the acoustic impedance of air and tissue are $42.8 \text{ g/cm}^2\text{s}$ and $1.6 \times 10^5 \text{ g/cm}^2\text{s}$ respectively. This difference is so large that most of the ultrasonic energy tends to be reflected at the interface. It is for this reason that a coupling medium like olive oil or special jelly is used to minimize the energy reflection by providing an air-free path between ultrasonic transducer and skin.



► **Fig. 23.1** Reflection and refraction of ultrasound at an interface between two media having different acoustic impedances

23.2.2 Wavelength and Frequency

Ultrasonics follow the general wavelength and frequency relationship given by

$$V = n\lambda$$

where V = propagation velocity of sound

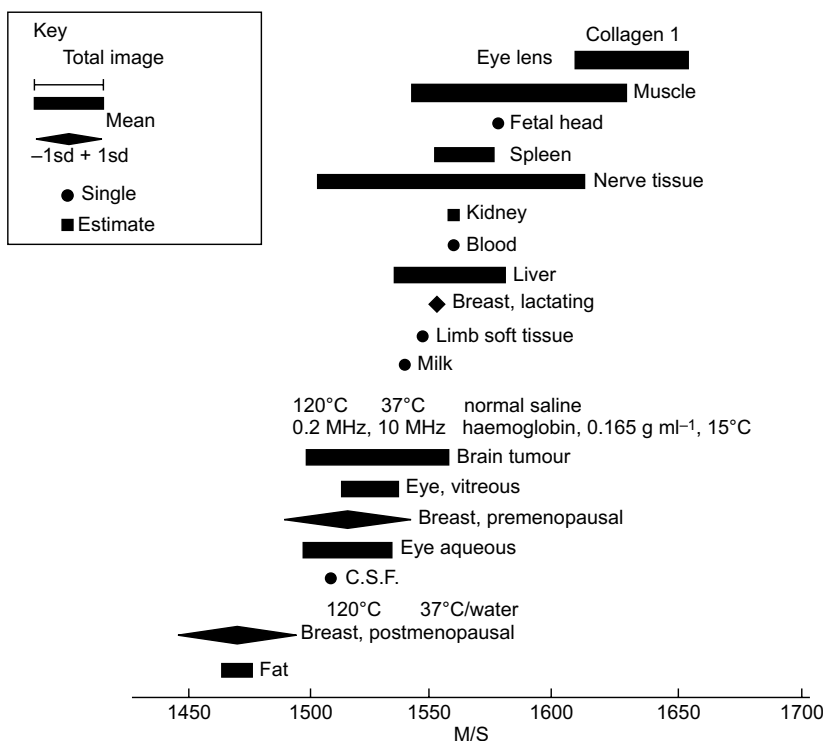
n = frequency or number of cycles which pass any given point in unit time

λ = wavelength, i.e., distance between any two corresponding points on consecutive cycles.

Ultrasonic frequencies employed for medical applications range from 1 to 15 MHz. This range also corresponds to radio frequencies (rf). However, there is an important basic difference between radio frequency and ultrasonic energy. Ultrasonic waves are transmitted as mechanical vibrations whereas rf energy would be in the form of electromagnetic radiations. No medium is necessary for propagation of rf energy and it would, therefore, pass even through vacuum; ultrasonic waves will, on the other hand, pass only through a medium.

23.2.3 Velocity of Propagation

Ultrasonic energy is transmitted through a medium as a wave motion and, therefore, no net movement of the medium is expected to occur. The velocity of propagation of the wave motion is determined by the density of the medium it is travelling through and the stiffness of the medium. At a given temperature and pressure, the density and stiffness of the biological substances are relatively constant, and, therefore, the sound velocity in them is also constant. The speed of ultrasound in m/s in various biological materials is shown in Fig. 23.2.



► **Fig. 23.2** Speed of ultrasound in various biological materials (Redrawn after P.N.T. Wells, *Biomedical Ultrasound*, Academic Press, London)

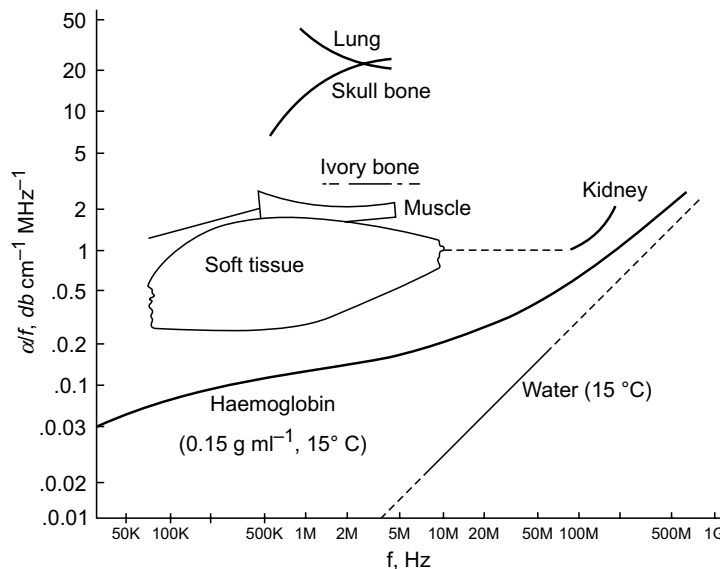
The knowledge of velocity of sound in a particular medium is important in calculating the depth to which the sound wave has penetrated before being reflected. If the time taken by the ultrasonic wave to move from its source through a medium, reflect from an interface and return to the source can be measured, then the depth of penetration is given by:

$$\text{Depth of penetration} = \frac{\text{Velocity of sound in the medium} \times \text{time}}{2}$$

The velocity of ultrasound in all body tissues is almost constant. Therefore, the depth of penetration can be read directly from the position of the echo pulse on the calibrated time axis of the oscilloscope trace.

23.2.4 Absorption of Ultrasonic Energy

The reduction of amplitude of ultrasonic beam while passing through a medium can be due to its absorption by the medium and its deviation from the parallel beam by reflection, refraction, scattering and diffraction etc. The relative intensity and the attenuation of an ultrasound beam is expressed in decibels (dB) and the absorption coefficient α is normally quoted in dB/cm. In soft tissues, α depends strongly on the frequency and therefore, for a given amount of energy loss, the lower frequency ultrasonic signal would travel more than the higher frequency signal. Quantitatively, the average value of sound absorption in soft tissues is of the order of 1 dB/cm/MHz. The attenuation of ultrasound in various biological materials (Wells, 1977) is shown in Fig. 23.3.



► Fig. 23.3 Attenuation of ultrasound in various biological materials (Adapted from P.N.T. Wells, *Biomedical Ultrasound*, Academic Press, London)

Table 23.1 shows the velocity of ultrasound, characteristic impedance and absorption coefficient in various materials.

• **Table 23.1** *Velocity of Ultrasound, Characteristic Impedance and Absorption Coefficient in Various Materials*

<i>Material</i>	<i>Velocity of ultrasound in m s⁻¹</i>	<i>Characteristic impedance, z in kg m⁻² s⁻¹ × 10⁻⁶</i>	<i>Absorption coefficient in dB cm⁻¹ at 1 MHz</i>
Air (20°C)	343	4×10^{-4}	12.0
Water	1480	1.48	0.002
Aluminium	6220	16.5	-
Fat	1450	1.38	0.6
Brain	1541	1.58	0.85
Liver	1549	1.65	0.9
Kidney	1561	1.62	1.0
Blood	1570	1.61	0.2
Muscle	1585	1.70	2.3
Skull-bone	4080	7.80	13.0
Lens of eye	1620	1.84	2.0
Human soft-tissue	1540	1.63	0.8

23.2.5 Beam Width

In general, ultrasonic waves are projected in a medium as a beam. Huygen's construction may be used to determine the spatial distribution of energy in this beam, which can be conveniently split into near and far fields. In the near field, within the first Fresnel zone, the beam is cylindrical with little spread. A series of maxima and minima is encountered in this region, as one travels out from the transducer which corresponds to constructive and destructive interference. The near field extends to a distance d from the transducer where

$$d = \frac{r^2}{\lambda} - \frac{\lambda}{4}$$

where r and λ are the radius of the transducer and the wavelength of the ultrasound respectively.

In the far field, the intensity of the beam reduces constantly with distance as it spreads out due to the finite size of the source. The angle of divergence within a cone of semi-angle θ about the central axis is given by

$$\begin{aligned} \theta &= \frac{0.61 \lambda}{r n} \text{ in radians} \\ &= \frac{0.61 V}{r n} = \frac{1.22 V}{n D} \end{aligned}$$

where n = frequency

V = velocity of sound waves

D = diameter of the transducer.

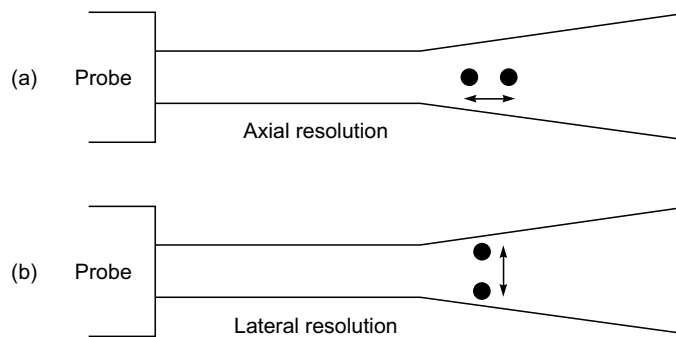
For example, a cm diameter transducer, excited at 1 MHz has a near field of about 10 cm in water and a semi-angle of divergence of 3.5 degrees. The beam shape may be modified by the use of focusing elements in front of the transducer.

23.2.6 Resolution

The resolution of an ultrasound system can be defined as the system's ability to distinguish between closely related structures. In general, the resolution is divided into axial and lateral resolution.

Axial Resolution: The axial resolution is the minimal axial distance [Fig. 23.4(a)], parallel to the beam axis, at which two reflecting structures are recognized as separate structures. The axial resolution is determined by the wavelength of the transmitted pulse. This means that the smaller the wavelength, the higher the frequency and better the axial resolution.

Lateral Resolution: The lateral resolution is the lateral distance, in a plane perpendicular to the beam axis, at which two reflecting structures can be seen as two separate structures [Fig. 23.4(b)]. The lateral resolution is determined by the shape/divergence of the ultrasound beam, produced by the probe.



► Fig. 23.4 (a) Axial resolution (b) lateral resolution

23.2.7 Generation and Detection of Ultrasound

The physical mechanism normally used to generate and detect ultrasonic waves is the piezo-electric effect exhibited by certain crystalline materials which have the property to develop electrical potentials on definite crystal surfaces when subjected to mechanical strain. The converse is also true, which means that mechanical displacement is produced when electrical charges are put on their surface. The effect is demonstrated by crystals of materials like quartz, tourmaline and Rochelle salt. This phenomenon offers an excellent method for converting electrical energy into mechanical energy and vice versa.

While working with natural crystals, it is difficult to establish the appropriate axis and cut the crystal in the required form. Therefore, quartz has generally been replaced by synthetic piezo- electric materials namely barium nitrate and lead zirconate titanate. They offer several advantages because they are far cheaper to produce and are much easier to construct transducers of complex shape and large areas. They can be moulded to any shape to obtain a better focusing action for producing high intensity ultrasonic waves.

The choice of piezo-electric material for a particular transducer depends upon its applications. Materials with high mechanical Q factor are suitable as transmitters whereas those with low

mechanical Q and high sensitivity are preferred as receivers and in case of non-resonance applications. Lead zirconate Titanate (PZT) crystals are much better than quartz crystals upto a frequency of about 15 MHz, because of its high electro-mechanical conversion efficiency and low intrinsic losses. The properties of PZT can be adjusted by modifying the ratio of zirconium and titanium and introducing small amounts of other substances such as lanthanum. PZT can operate at temperatures up to 100°C or higher and it is stable over long periods of time. It is mechanically strong and can be machined to various shapes and sizes. At frequencies higher than this, quartz is normally used because of its better mechanical properties. Polyvinylidene difluoride (PVDF) is another ferro-electric polymer that has been used effectively in high frequency transducers. The surface of the synthetic crystals is normally silvered for making external electrical connections.

Piezo-electric crystals are available in several shapes and the selection of a particular shape depends upon the application to which it is to be put. The application of a voltage to the transducer disc causes changes in its thickness, thereby giving rise to longitudinal waves propagated along the axis perpendicular to its face. The magnitude of the waves will be a maximum at the resonant frequency of the disc, which is determined by its thickness. Likewise, when the transducer is acting as a receiver, it will be most sensitive to ultrasonic vibrations at its own resonant frequency. Therefore, considerable attention is devoted to tailoring the dimensions of the transducer to the frequency at which it will be used.

There are three parameters that are important in optimizing transducers for various types of applications (Hunt et al. 1983). These are frequency, active element diameter and focusing. Their effects on the performance are as follows:

Frequency: With increase in frequency, the sound beam becomes more directional and the axial resolution improves. However, due to attenuation of higher frequency ultrasound waves in the tissues, the penetration decreases. For most abdominal ultrasound examinations, the frequencies used are in the range of 1-5 MHz, whereas the wavelength is in the range of 1 mm. Higher frequencies (10-15 MHz) are used for superficial organs, such as the eye, where deep penetration is not required and where advantage may be taken of the 0.1 mm wavelength to improve geometrical resolution.

The following are general rules which apply to frequency:

↑ Frequency	↑ Axial Resolution	↑ Lateral Resolution	↓ Penetration
↓ Frequency	↓ Axial Resolution	↓ Lateral Resolution	↑ Penetration

Frequency also influences lateral resolution by affecting beam divergence. The following rule applies, assuming all other factors remain constant.

↑ Frequency	↓ Beam divergence	↑ Lateral Resolution
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Active Element Diameter (AED): As the transducer face diameter increases, the beam width decreases and therefore, lateral resolution improves.

↑ A.E.D.	↓ Rate of Divergence	↑ Lateral Resolution
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The choice of which element size to use is generally based on two considerations: where on the patients' body the transducer is to be positioned, and the depth in the body to the structures of interest. While it is often desirable to use a large diameter to reduce beam divergence, this is sometimes not practical. For example, many times in echocardiography, a patient will have

very narrow inter-costal spaces, necessitating the use of small diameters. In thyroid applications, the area to be scanned is irregularly shaped. Consequently, small diameters are helpful in maintaining good contact between the patient and the transducer. General abdominal and pelvic examinations usually do not have restrictions such as these, and large diameters can be used easily.

The second point to consider is the depth in the body you wish to image. For the best image detail, it is advantageous to have the ultrasonic beam as narrow as possible (minimum divergence). For superficial structures, the beam must narrow very close to the transducer face and lateral resolution at greater tissue depths is not critical; therefore, small diameter transducers are indicated. When the structures of interest lie deeper in the body (pelvic and obstetrical examinations, obese patients) large diameters are advised, since their decreased rate of beam divergence becomes important at greater tissue depths.

Focusing: Focusing a transducer is a mean of minimizing the beam width and adjusting the focal zone to give optimum results for a particular examination. Acoustic lenses can be used to shape the ultrasonic beam pattern. The width of the beam can be made narrow with the result that better lateral resolution can be obtained. The focal point can be selected at different depths from the face of the transducer. The ability to select different focal points allows for the optimization of transducers for a particular type of studies. Modern transducers are internally focused and externally are of flat face.

While selecting a transducer for a particular application, all three of these factors should be carefully weighed to achieve optimum performance.

23.3 MEDICAL ULTRASOUND

The use of ultrasound in the medical field can be divided into two major areas: the therapeutic and the diagnostic. The major difference between the two applications is the ultrasonic power level at which the equipment operates. In therapeutic applications, the systems operate at ultrasonic power levels of upto several watts per square centimetre while the diagnostic equipment operates at power levels of well below 100 mW/cm^2 . The therapeutic equipment is designed to agitate the tissue to the level where thermal heating occurs in the tissue, and experimentally has been found to be quite successful in its effects for the treatment of muscular ailments such as lumbago.

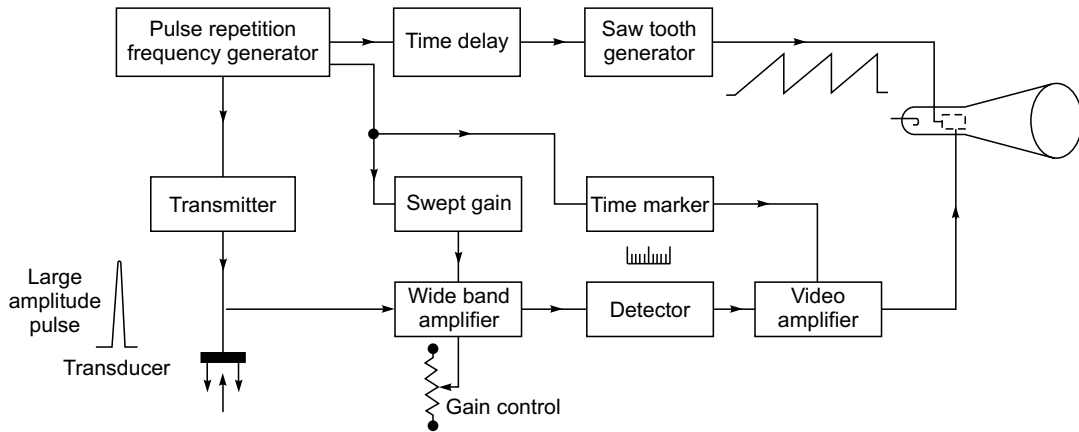
For diagnostic purposes, on the other hand, as long as a sufficient amount of signal has returned for electronic processing, no additional energy is necessary. Therefore, considerably lower ultrasonic power levels are employed for diagnostic applications. Since the absorption of ultrasound in tissue is proportional to the operating frequency which, in turn, is related to the desired resolution (ability to detect a certain size target) the choice of ultrasonic power level used is often dictated by the application. Diagnostic ultrasound is used either as continuous waves or in the pulsed wave mode. Applications making use of continuous waves depend for their action on Doppler's effect. Among the important commercially available instruments based on this effect are the foetal heart detector and blood flow measuring instruments.

There are, however, many applications where only pulsed waves can be employed. In fact, the majority of modern ultrasonic diagnostic instrumentation is based on the pulse-technique. Pulse-echo based equipment is used for the detection and location of defects or abnormalities in the structures at various depths in the body. This is possible because the time of travel of a short pulse can be measured with much greater ease as compared to continuous waves.

Echoencephalograph, echocardiograph and ultrasonic scanners for imaging are all based for their working on the pulse technique.

23.4 BASIC PULSE-ECHO APPARATUS

Pulse-echo technique of using ultrasound for diagnostic purposes in medical field was first attempted by making use of flaw detectors normally employed in industry for non-destructive testing of metallic structures. The basic layout of the apparatus based on this principle is shown in Fig. 23.5.



► Fig. 23.5 Block diagram of a basic pulse-echo system

The pulse-echo technique, basically, consists in transmitting a train of short duration ultrasonic pulses into the body and detecting the energy reflected by a surface or boundary separating two media of different specific acoustic impedances. With this technique, the presence of a discontinuity can be conveniently established and its position located if the velocity of travel of ultrasound in the medium is known. Also, it is possible to determine the magnitude of the discontinuity and to assess its physical size.

The transmitter generates a train of short duration pulses at a repetition frequency determined by the PRF generator. These are converted into corresponding pulses of ultrasonic waves by a piezo-electric crystal acting as the transmitting transducer. The echoes from the target or discontinuity are picked up by the same transducer and amplified suitably for display on a cathode ray tube. The X plates of the CRT are driven by the time base which starts at the instant when the transmitter radiates a pulse. In this way, the position of the echo along the trace is proportional to the time taken for a pulse to travel from the transmitter to the discontinuity and back again. Knowing the velocity of ultrasonic waves and the speed of the horizontal movement of trace on the CRT, the distance of the target from the transmitting end can be estimated.

The Probe: The transducer consists of a piezo-electric crystal which generates and detects ultrasonic pulses. The piezo-electric materials generally used are barium titanate and lead zirconate titanate. The crystal is cut in such a way that it is mechanically resonant of an increased efficiency of conversion of electrical energy to acoustic energy. It is usually one half wavelength thick for the particular frequency used.

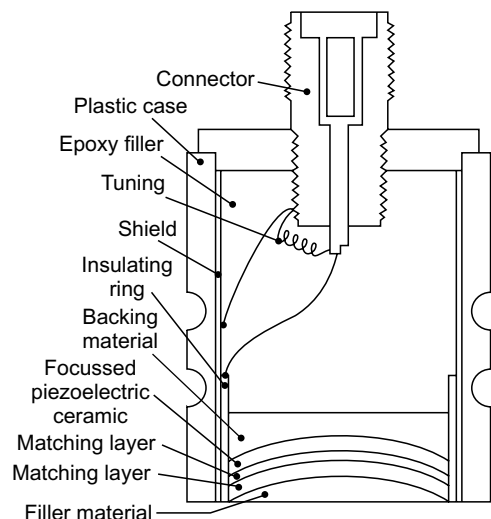
When the transducer is excited at its resonance frequency, it will continue to vibrate mechanically for some time after the electrical signal ceases. This effect is known as 'after ringing' and destroys the precision with which the emission or detection of a signal can be timed. To reduce it, the transducer must have a good transient response and consequently a low Q is desirable. To achieve this, the transducers are normally damped. This can be done by controlling the rear surface to have a high impedance and high absorbancy of ultrasonic waves. This is to ensure that the energy radiated into it does not return to the transducer to give rise to spurious echoes. Backing material, therefore, is an important consideration in transducer construction. It is generally an epoxy resin loaded with a mixture of tungsten and aluminium powder. Backing material is made thick enough for complete absorption of the backward transmitted ultrasonic waves.

The probes are designed to achieve the highest sensitivity and penetration, optimum focal characteristics and the best possible resolution. This requires that the acoustic energy be transmitted efficiently into the patient. It is thus desirable to reduce the amount of reflected acoustic energy at the transducer-body interface. The single quarter wavelength matching layer accomplishes this by interposing a carefully chosen layer of material between the transducer's piezo-electric element and body tissue. A material with an acoustic impedance between tissue and piezo-electric ceramic is selected to reduce the level of acoustic mismatch at the transducer body interface. A uniform thickness of one-quarter wavelength for a frequency at or near the transducer's centre frequency results in higher acoustic transmission levels because of the favourable phase reversals within the layer.

The single quarter wavelength design, however, provides optimal transmission of ultrasonic energy at a particular wavelength only. This presents a problem for diagnostic pulse-echo ultrasound which is characterized by very short pulses containing a broad band of frequencies.

Also, the single quarter wavelength matching layer transducer has a face with a concave curvature. Occasionally, this can lead to air bubble entrapment or patient contact problems. Multilayer matching (Fig. 23.6) technology overcomes these problems by interposing two layers between the piezo-electric element and body. Two materials are chosen with acoustic impedances between the values for ceramic and tissue. A stepwise transition of impedance from about 30 for ceramic to about 1.5 for tissue allows even further reduction of this acoustic impedance mismatch. The concavity can be filled with a material which is as acoustically transparent as possible, thus yielding a transducer with a hard, flat face for good patient contact, while minimally affecting the ultrasound beam.

Pulse Repetition Frequency Generator: This unit produces a train of pulses which control the sequence of events in the rest of the equipment. The PRF is usually kept between 500 Hz to 3 kHz. There are several standard circuits for producing the required type of waveforms. These circuits could be the blocking oscillator or some form of the astable multivibrator. Mostly, the latter is preferred

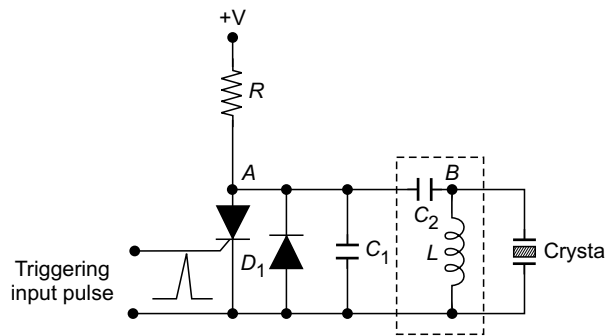


► **Fig. 23.6** Multilayer matching between transducer's piezoelectric element and body tissue (Courtesy: K-B Aerotech, U.S.A.)

because the pulse duration can be more conveniently varied and the circuit does not require the use of a pulse transformer.

The width of the output pulse from the PRF generator should be very small, preferably of the order of a micro-second, to generate short duration ultrasonic pulse. It is a practice to use one astable circuit to generate a train of pulses with the required frequency and then to use them to trigger a monostable multivibrator which produces pulses of the required width. With the short pulse duration and the repetition rate of 1 kHz, only a few micro-seconds are occupied by the emission of the pulse, and the transducer is free to act as a receiver for the remainder of the time.

Transmitter: The transmitting crystal is driven by a pulse from the PRF generator and is made to trigger an SCR circuit which discharges a capacitor through the piezo-electric crystal in the probe to generate an ultrasonic signal. The circuit typically employed is shown in Fig. 23.7.



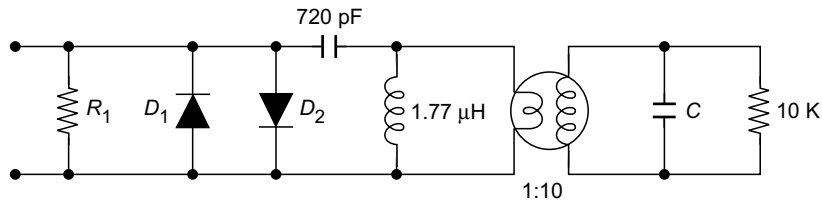
► Fig. 23.7 Circuit diagram of a transmitter used in pulse-echo application

Under normal conditions, the SCR is non-conducting. The capacitor C_1 can charge through the resistance R to the $+V$ potential. If a short triggering positive pulse is applied to the gate of the SCR, it will fire and conduct for a short time. Consequently, the voltage at 'A' will fall rapidly resulting in a short duration, high voltage pulse at 'B'. This pulse appears across the crystal which generates short duration ultrasonic pulse. For producing a pulse with a very short duration it is necessary to use an SCR with a fast turn 'on' time and high switching current capability, which can be able to withstand the required supply voltage. SCR 2N4203 can be used because of its high peak forward blocking voltage (700 V), high switching current capability (100 A) and fast turn-on time (100 ns).

Receiver: The function of the receiver is to obtain the signal from the transducer and to extract from it the best possible representation of an echo pattern. To avoid significant worsening of the axial resolution, the receiver bandwidth is about twice the effective transducer bandwidth.

Transmitter-Receiver Matching: Ultrasonic pulse-echo systems generally use the same transducer crystal for both transmitting the ultrasonic energy and receiving the reflected echo. This permits the systems to have a compact transducer and also produces a symmetrical and well-defined beam shape. However, using a common source-receiver of ultrasound means that the sensitive input stage of the receiving amplifier must be protected from the high voltage transmission pulse. Such protection is usually provided by using a circuit shown in Fig. 23.8 (Follett and Ackinson, 1976).

The tuned transducer and input matching transformer form part of a Butterworth filter with Q of about 1.4 to reject transducer ringing. On transmission, the diodes conduct, protecting the



► Fig. 23.8 Circuit for input matching of transmitter-receiver (after Follett and Ackinson, 1976)

receiver in conjunction with L_2 and effectively connecting L_2 in parallel with L_1 . Since $L_2 = 4L_1$, this has negligible effect on the transducer tuning due to the low circuit Q .

In the modern ultrasound imaging systems, a digital beamformer typically generates the necessary digital signals with the proper timing and phase to produce a focused transmit signal. The beamformer generates complex transmit waveforms using an arbitrary waveform generator to optimize image quality. It generates digital 8-bit to 10-bit words at rates of approximately 40 MHz to produce the required transmit waveform. Digital-to-analog converters (DACs) are used to translate the digital waveform to an analog signal, which is then amplified by a linear high-voltage amplifier to drive the transducer elements. This transmit technique is generally reserved for more expensive and less portable systems, as it can be very large, costly, and power hungry. As a result, the majority of ultrasound systems do not use this transmit-beamformer technique, but instead use multilevel high-voltage pulsers to generate the necessary transmit signals.

In this alternate implementation highly-integrated, high-voltage pulsers quickly switch the transducer element to the appropriate programmable high-voltage supplies to generate the transmit waveform. To generate a simple bipolar transmit waveform, a transmit pulser alternately connects the element to a positive and negative transmit supply voltage controlled by the digital beamformer.

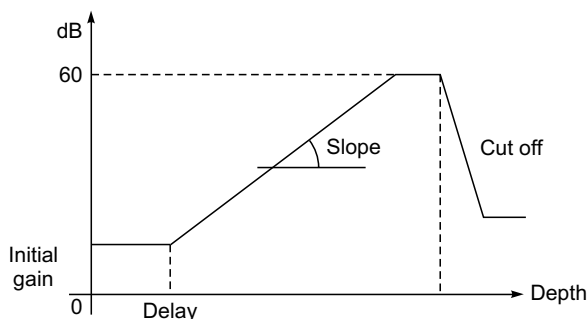
Wide Band Amplifier: The echo-signals received at the receiving transducer are in the form of modulated carrier frequency and may be as small as a few microvolts. These signals require sufficient amplification before being fed to a detector circuit for extracting modulating signals which carry the useful information. This is achieved in a wide-band amplifier, which is wide enough to faithfully reproduce the received echoes and to permit the use of different transducers operating at several different frequencies. A desirable gain of wide band amplifier is of the order of 80-100 dB. It must also have a very wide dynamic range so that the amplifier does not operate in the non-linear regions with large input signals. The amplifier must also have a low noise level to receive echoes from deep targets. The input amplifier is usually a dual gate MOSFET which is very suitable for high frequency signals and provides a high input impedance to the signals from the transducer.

Due to the wide dynamic range of echo-amplitudes that are contained in an ultrasonic image, a log amplifier is usually utilized. In this amplifier, the output voltage is proportional to the logarithm of the input voltage. By utilizing a log amplifier, one can see small relative differences in both low amplitude and high amplitude echoes in the same image.

Swept Gain Control: Stronger echoes are received from the more proximal zones under examination than from the deeper structures. The receiving amplifier can only accept a limited range of input signals without overloading and distortion. Abrupt changes in tissue properties that shift the acoustical impedance can cause the echo amplitudes to vary over a wide dynamic

range, perhaps 40 to 60 dB. In order to avoid this, the amplifier gain is adjusted to compensate for these variations. This reduces the amplification for the first few centimetres of body tissue and progressively increases it to a maximum for the weaker echoes from the distal zone. In some instruments, segmented gain control arrangements are made to control the gain in segments on the time axis. This permits one to selectively amplify or reject echoes from different structures located at different depths. The swept gain profile can be adjusted and displayed on most of the modern instruments.

A simple technique (Fig. 23.9) of providing tissue attenuation compensation is to include a basic receiver gain control to echoes from near the skin surface. A typical control range is 0 to 60 dB. The rate of gain increase with depth is then set by a 'slope' or 'rate control' which typically provides for 0 to 10 dB/cm gain increase. The maximum gain value is reached by a 'far gain' control, also commonly with a 0 to 60 dB range (Maginness, 1979).



► Fig. 23.9 A simple technique of providing tissue attenuation compensation in ultrasound pulse echo systems.

Detector: After the logarithmic amplification, the echo signals are rectified in the detector circuit. The detector employed could be of the conventional diode-capacitor type with an inductive filter to have additional filtering of the carrier frequency. In this rectification process, the negative half-cycles in the echo voltage waveforms are converted into positive half-cycles. This is followed by a demodulation circuit in which the fundamental frequency signal upon which the echo amplitude information has been riding, is eliminated. The output of the demodulator circuit is in the form of an envelope of the echo signal. The conventional demodulators which work well with low modulation frequencies give a much poorer performance with pulses containing only a few carrier cycles. With a peak input of 5 V, it is difficult to exceed a dynamic range of 25-30 dB. Much wider ranges of up to 40 dB can be obtained by cascade demodulation in which the individual demodulator characteristics are superimposed to cover different input signal amplitude ranges.

The output of the demodulator is the information desired, i.e. the amplitude of the echo signal and its time delay from the transmission pulse. The magnitude of the echo amplitude information can be controlled as it is written into the ultrasonic image. It is a great advantage to have this control due to the large variability of patient anatomy and acoustic parameters of normal tissue.

Video Amplifier: The signal requires further amplification after its demodulation in the detector circuit before it can be given to the Y-plates of the Cathode ray tube (CRT). The output of the detector circuit is typically around 1 V, but for display on the CRT, the signal must be amplified to about 100 to 150 V. In addition to this, the amplifier must have a good transient response with minimum possible overshoot. The most commonly used video amplifier is the RC coupled type, having an inductance in series with the collector load. The inductance helps in extending the high frequency response of the amplifier.

The video signal is subject to some kind of compression function. This involves compression of echoes in the mid-amplitude range, denoting relatively larger sections of the output range to

very weak and very strong echoes. This feature is useful in maintaining gain for weak echoes while avoiding saturation on strong echoes. With the use of digital scan converters in ultrasound imaging system, mid-range expand, mid-range compress, logarithmic and log-linear amplitude transfer functions can be made operator selectable.

The video signal representing the echo envelope is subjected to another processing step for 'edge enhancement'. It is obtained by partially differentiating the signal. The normal practice is to add a small fraction of the differentiated signal to the original signal. The added fraction is kept variable with an 'echo process' or 'enhancement control'.

Time Delay Unit: The time delay unit is sometimes required for special applications. Normally, the time base will begin to move the spot across the CRT face at the same moment as the SCR (Fig. 23.7) is fired. If desired, in special cases, the start of the trace can be delayed by the time delay unit so that the trace can be expanded to obtain better display and examination of a distant echo.

Time Base: The time base speed is adjusted so that echoes from the deepest structures of interest will appear on the screen before the beam has completely traversed it. Taking the speed of ultrasound in soft tissue to be about 1,500 m/s, a time of 13.3 μ s must be allowed for each centimetre that the reflecting interface is below the surface. In many applications, distance markers appropriate to each time-base setting appear directly on the screen, which greatly simplifies distance measurements. Several standard circuits are available for generating the sawtooth waveform to provide a time base suitable for horizontal deflection of the spot on the CRT screen. The horizontal sweep generator is controlled by the PRF generator as the sweep starts at the moment that the transmitting pulse is applied to the transducer.

Time Marker: The time marker produces pulses that are a known time apart and, therefore, correspond to a known distance apart in human tissues. These marker pulses are given to the video amplifier and then to the Y plates for display along with the echoes.

Display: After amplification in the video amplifier, the signal is given to the Y plates of the CRT. CRT is not only a fast-acting device but also gives a clear presentation of the received echo signals.

23.5 IMAGING MODES

Various imaging modes found in today's ultrasound systems are:

- *A-mode (Amplitude)* imaging displays the amplitude of a sampled voltage signal for a single sound wave as a function of time. This mode is considered 1D and used to measure the distance between two objects by dividing the speed of sound by half of the measured time between the peaks in the A-mode plot, which represents the two objects in question.
- *M-mode (Motion)* display refers to scanning a single line in the object and then displaying the resulting amplitudes successively. This shows the movement of a structure such as heart. Because of its high pulse frequency (up to 1000 pulses per second), this is useful in assessing rates and motion and is used extensively in cardiac and foetal cardiac imaging.
- *B-mode (Brightness)* imaging is the same as A-mode, except that brightness is used to represent the amplitude of the sampled signal. B mode imaging is performed by sweeping the transmitted sound wave over the plane to produce a 2D image. Typically, multiple sets of pulses are generated to produce sound waves for each scan line, each set of pulses are intended for a unique focal point along the scan line.

- *CW (Continuous Wave) Doppler*: Here, Doppler, a sound wave at a single frequency is continuously transmitted from one piezo-electric element and a second piezo-electric element is used to continuously record the reflected sound wave. By continuously recording the received signal, there is no aliasing in the received signal. Using this signal, the blood flow in veins can be estimated using the Doppler frequency. However, since the sensor is continuously receiving data from various depths, the velocity location cannot be determined.
- *PW (Pulsed Wave) Doppler*: In this mode, several pulses are transmitted along each scan line and the Doppler frequency is estimated from the relative time between the received signals. Since pulses are used for the signaling, the velocity location can also be determined.
- *Colour Doppler*: the PW Doppler is used to create a colour image that is super-imposed on top of B-mode image. A colour code is used to denote the direction and magnitude of the flow. Red typically denotes flow towards the transducer and blue denotes flow away from it. A darker colour usually denotes a larger magnitude while a lighter colour denotes a smaller magnitude.
- *Harmonic Imaging*: It is a new modality where the B-mode imaging is performed on the second (or possibly other) harmonics of the imaging. Due to the usual high frequency of the harmonic, these images have higher resolution than conventional imaging. However, due to higher loss, the depth of imaging is limited. Some modern ultrasound systems switch between harmonic and conventional imaging based on depth of scanning. This system imposes stringent linearity requirements on the signal chain components.

23.5.1 A-Scan

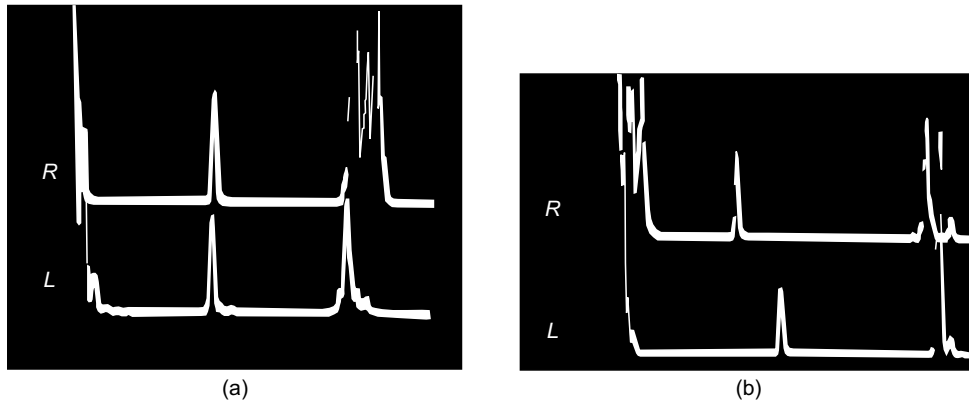
This type of scan offers only one-dimensional information. The echo signals are applied to the Y-deflecting plates of the CRT so that they are displayed as vertical blips as the beam is swept across the CRT. The height of the vertical blip corresponds to the strength of the echo and its position from left to right across the CRT face corresponds to the depth of its point of origin from the transducer.

23.5.1.1 Applications of A-Scan

Echoencephalograph: In the normal brain, the mid-line surfaces are parallel to the flat areas of the bone near the ear. When there is a head injury, the brain gets tilted to one side or the other due to bleeding, but it still retains its normal shape. In such cases, the echoes can be easily obtained but they are placed at different distances from the probe, when the probe is placed first on one side and then on the other side of the skull (Fig. 23.10).

Even when a tumour grows in the brain, the anatomy of the brain is gently altered and there is usually considerable tilting and displacement of the brain ventricles. Ultrasonic mid-line echo, in such cases, immediately establishes the abnormality of the brain due to its shift to one side from the centre.

The instrument for diagnosis and detection of the mid-line of the brain is called 'echoencephaloscope'. It usually incorporates a measuring range of 0–18 cm of tissue depth. The normally used frequency range is 1–3 MHz. The probe for 2 MHz with the diameter of 15–20 mm is the most common and gives a good resolution. The probe for a 1 MHz allows for deep penetration and may preferably be employed for elderly patients whose skulls are strongly calcified.



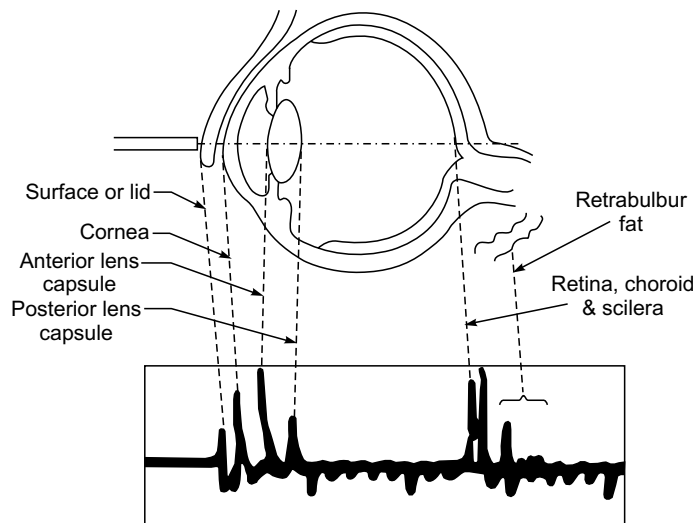
► **Fig. 23.10** Echoes received from the brain (a) in the normal brain, the mid-line echoes coincide for each way (b) in the abnormal case, there is a shift between the two echoes

23.5.1.2 Echo-ophthalmoscope

A-mode ultrasonic technique was found to be useful in ophthalmology for the diagnosis of retinal detachments, intra-ocular tumours, vitreous opacities, orbital tumours, and lens dislocation. It helps in the measurement of axial length in patients with progressive myopia, localization of intra-ocular foreign bodies and extraction of non-magnetic foreign bodies.

Echo-ophthalmoscopy employs a 7.5–15 MHz pencil type transducer. The transmitted pulse should be of very small width (in nanosec) and range.

In the normal eye, as is shown in Fig. 23.11, echoes can be obtained from the following structures when performing an anterior-posterior examination along the optical axis: the surface of the lid (corresponding with the zero marker that appears constantly on the trace); the surface



► **Fig. 23.11** Acoustic landmarks in the normal eye along the optical axis. The electronic markers shown on the figure represent 2 mm of actual tissue depth

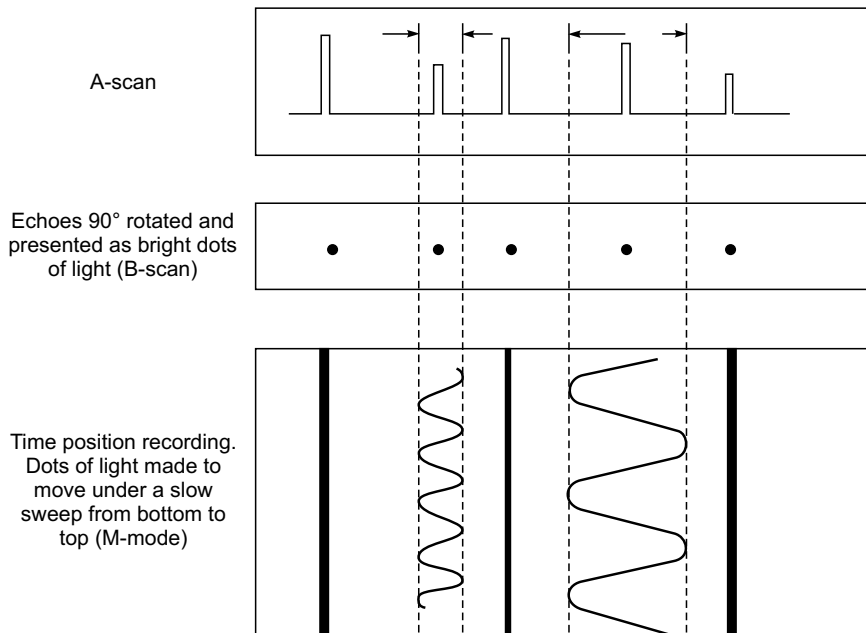
of the cornea; the anterior lens capsule, the posterior lens capsule, the posterior wall of the globe (usually appearing as a complex of echoes originating from the retina, choroid and sclera as a unit) and retro-bulbar fat.

By using the distance markers on the trace, the depth and relative position of each structure represented in the display can be read directly from the screen or the resulting photograph. For example, in Fig. 23.11, the corneal echo appears at 2 mm and the posterior wall echo at 26 mm, giving a gross axial length measurement of 24 mm. In this eye, the depth of the anterior chamber and the thickness of the lens are each shown to be approximately 3 mm.

A-mode is no longer used in the modern ultrasound systems.

23.5.2 M-Mode (Echocardiograph)

In A-mode display, ultrasonic echoes produce vertical displacements of a horizontal trace on CRT. Basically, the amount of vertical displacement is proportional to the strength of the echo, and the distance along the horizontal trace represents the time of sound travel in human tissue, thereby permitting accurate measurement of tissue depth between any two echo sources. By electronically rotating the A-mode echoes 90° towards the viewer, the echoes can be presented as bright dots of light along an imaginary horizontal base line as shown in Fig. 23.12. The distance between the dots again represents time or tissue depth and the intensity of the dots represents the strength of the echoes.

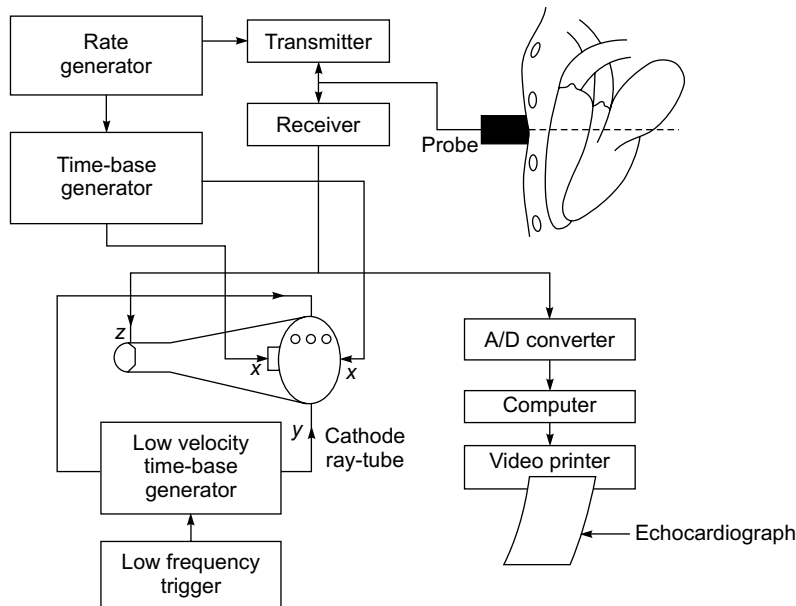


► Fig. 23.12 Principle of time-motion (M-mode) display

If one of the echo sources is a moving structure, then the echo dots of light from that structure will also move back and forth. If the dots are made to move with an electronic sweep, from bottom to the top of the screen at a pre-selected rate of speed, the moving dots will trace out the motion pattern of the moving structure. This display is known as M-mode display. If a

photographic film is continuously exposed to one sweep cycle of this display, a composite picture will result, providing a waveform representation of the motion pattern of the moving structure. Alternately, thermal video printers are used for recording the M-mode information.

Fig. 23.13 shows the block diagram of an echocardiograph, which is based on the M-mode. Several circuit blocks are common to the general echo measuring instrument, except for the addition of a slow sweep circuit and recording arrangement.

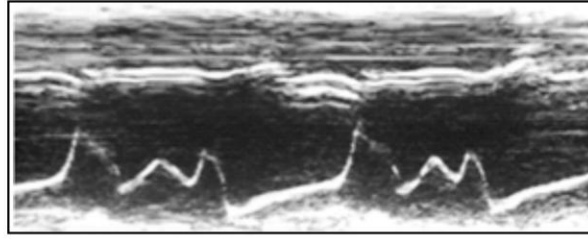


► Fig. 23.13 Block diagram of an echocardiograph circuit

For echocardiography, the transducer is placed between the third and the fourth ribs on the outer chest wall where there is no lung between the skin and the heart. From this probe, a low intensity ultrasonic beam is directed towards the heart area and echo signals are obtained. The probe position is manipulated to obtain echoes from areas of interest in the heart.

Echocardiograph is a widely used and valuable instrument for carrying out cardiac examination and assessment of many congenital and acquired cardiac diseases. By using this instrument it is possible to detect intra-cardiac structures. The movement of these structures can also be recorded with the better resolution than with angiographic diagnostic technique. The instrument presents time-versus-motion information about heart structures on slow speeds normally used in electrocardiogram recordings. When an ECG trace is superimposed on the ultrasonic display, the movement of structures detected ultrasonically can be conveniently correlated with known events in the cardiac cycle. Phonocardiogram is also often recorded simultaneously.

The echocardiogram is currently the best method for the diagnosis of mitral stenosis. Echocardiography is also often used for the study of the aortic valve, tricuspid valve and pulmonary valve. Another very important use is in the detection of pericardial effusion, which is the abnormal collection of fluid between the heart and the pericardial sac. These examinations can be performed quickly and easily at the bedside, if necessary, without apparent risk to the patient. Fig. 23.14 shows a typical echocardiogram.



► Fig. 23.14 Typical echocardiogram – Mitral valve motion (www.drsvenkatesan.co.in)

Pulsed Doppler echocardiography (Baker *et al.*, 1976) depends on the sensing of blood flow velocity in contrast to M-mode echocardiography which is based on the anatomical (dimensional) properties of the heart. The technique is used as an adjunct to conventional M-mode echocardiography and frequently information obtained from the pulsed Doppler examination compliments or reinforces the M-mode procedure. In many cases, pulsed Doppler findings provide useful diagnostic information whereas M-mode findings may be nearly normal or suggestive.

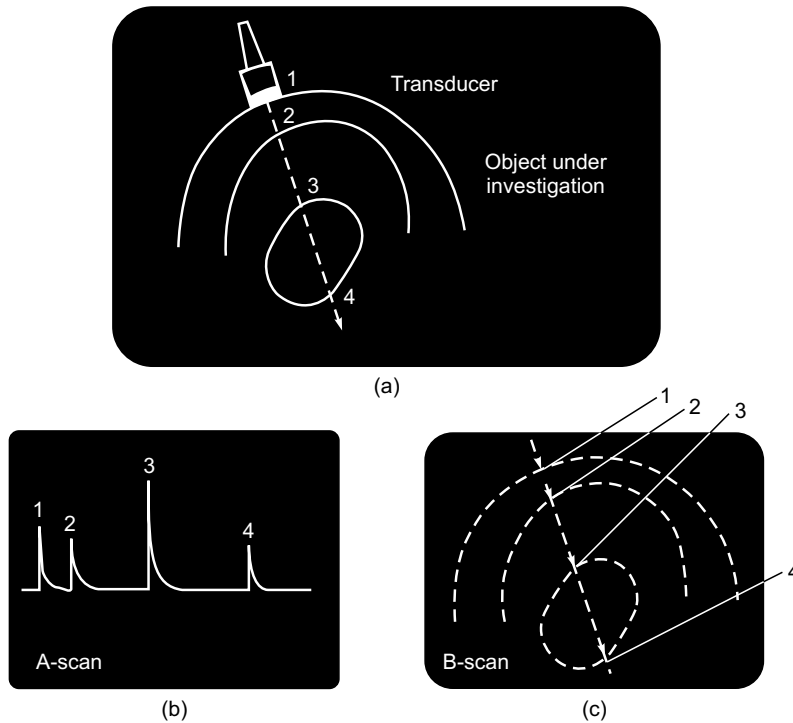
The system operates on the principle of reflected ultrasound and senses flow velocity within a small 2×4 mm tear-drop shaped volume, referred to as the sample volume. The sample volume is specifically selectable within the heart and great vessels by means of a depth control setting and is subject to a variety of components of blood flow velocity; laminar, turbulent and motional components like wall motion and valve motion. These components are isolated by appropriate filters in the circuitry and each has associated audio-tonal qualities and spectral patterns.

23.5.3 B-Mode (B-Scanner)

Obviously the A-scope display is very difficult to interpret when many echoes are present simultaneously and often potentially useful information is wasted. A pictorial display can be conceived as a means of simultaneously presenting the echo information as well as information about the position of the probe and the direction of propagation of the sound. This is achieved in the B-scan display which results from brightness modulation with amplitude of the echoes obtained for various probe positions and orientations to produce a cross-sectional image of the object integrated by a storage display from individual scans.

Fig. 23.15 represents the difference between A-scan and B-scan. Fig. 23.15 (a) is a hypothetical body with the probe placed upon its surface. The probe is positioned in such a way that it transmits a beam obliquely inwards. The beam encounters three interfaces in its travel. A-scan representation [Fig. 23.15 (b)] of this structure consists of vertical peaks (2, 3, 4) received as echoes at the receiving crystal in response to the transmitted pulse (1). The same structure in B-scan [Fig. 23.15 (c)] appears as light dots whose position is related to the echoing interface within the body.

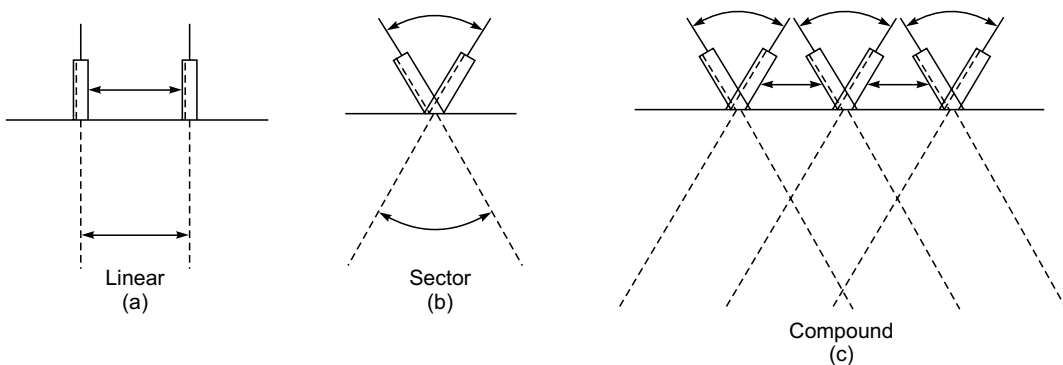
In order to record cross-sectional pictures of internal structures, the ultrasonic probe is mounted on a mechanical scanner which allows movement in two directions and which links the direction and position of a B-scope time base on a CRT to those of the ultrasonic beam within the patient.



► Fig. 23.15 Difference between A-scan and B-scan displays

23.5.3.1 Types of Scans

Three types of scanning arrangements are utilized for building cross-sectional images using ultrasound. They are illustrated in Fig. 23.16. The most common scan used for abdominal studies is the linear scan [Fig. 23.16(a)]. A linear scan is when the ultrasonic transducer remains parallel to the surface of the object being examined and the sound beam is perpendicular to the transducer movement. Only the location of the transducer is changed but the angle of the beam is



► Fig. 23.16 Types of scanning arrangement (a) linear scan, (b) sector scan, (c) compound scan

held constant. The most common scan used in echocardiography is the sector scan [Fig. 23.16(b)]. The scan is made by rocking the transducer about a fixed point such that the sound beam covers a sector. Compound scanning [Fig. 23.16(c)] is merely a combination of linear and sector scans.

23.5.4 B-scanner Imaging Instrumentation

Modern ultrasonic imaging equipment is quite complex, with different manufacturers using different signal processing techniques in their equipment and offering different types of displays. However, all ultrasonic imaging systems contain the same basic building block circuits, namely: transmitter, receiver, memory and display. The transmitter and receiver circuits used in the imaging systems are similar to those which are explained under the basic pulse-echo system. However, in the case of imaging systems, the output of the video amplifier is given to the memory for scan conversion and subsequent display on a LCD monitor.

B-scan is based on the fundamental information provided by echoes in an A-scan mode which is used to modulate intensity of the CRT electron beam instead of deflecting it vertically. The transducer is moved manually for scanning the region of interest on the body. For obtaining a 2-dimensional cross-section image (B-scan), it is necessary to know the transducer position and its orientation. The transducer is coupled by shaft encoders and position sensors to the co-ordinate generator so as to correlate the origin of the echoes from various structures of the body on the CRT scan.

B-scanning of static objects gives two-dimensional images which allow assessment of size, shape and position of the examined structures. If, however, the object is moving, the quality of the picture is degraded, the degree of degradation being proportional to the range and velocity of the movement. Therefore, fast working real time scanners have been developed to study both static and dynamic structures in the human body.

23.6 REAL-TIME ULTRASONIC IMAGING SYSTEMS

One of the serious limiting factors in B-scanning is the length of time taken to complete a scan. This results in blurring and distortion of the image due to organ movement, as well as being tedious for the operator. Elimination of motion artifact is important in conventional B-scanning but is critical if rapidly moving areas such as the chambers of the heart are to be made visible. Rapid scanning techniques have been developed to meet these needs. The approaches used include fast physical movement of a single transducer. Alternatively, electronic methods using arrays of transducers which can be triggered in sequence or in groups, may be utilized. In these systems, electronic manipulation allows the beam to be swept rapidly through the area of interest. Finally, instruments in which an array of transducers is combined with mechanized motion, have also been introduced. These instruments are called real-time scanners as there is negligible time delay between the input of data and the output of processed data in such systems.

A very important property of ultrasonography is the short image reconstruction time of about 20 to 100 ms, permitting the real-time scanning and observation of processes in the organs of the body. Consequently, sonography is well-suited to the fast, interactive screening of even larger organ regions and the display of dynamic processes, such as cardiology. Ultrasound section scans are, by contrast with CT and MR, largely free of kinetic blurring or motion artifacts (Haerten, 1994). On the other hand, ultrasound images are only section scans, and not whole body scans.

The real-time systems, therefore, have the following characteristics:

- Possibility of studying structure in motion—this is important for cardiac and foetal structures.
- During observation, the scan plane can be easily selected since the echo image appears instantaneously on the display.

23.6.1 Requirements of Real Time Ultrasonic Imaging Systems

The primary requirements of an ultrasonic imaging system are: high resolution, long range, adequate field of view, sufficiently high frame rate and high detectivity. However, high *resolution* or the ability of the system to resolve fine spatial dimensions is a key performance requirement. A resolution of 1–3 mm in all three spatial dimensions is desirable for a number of diagnostic studies like the early detection of tumours or other pathological conditions. The depth resolution (resolution along the axis) is usually governed by transducer pulse width and subsequent filtering. The round trip propagation time over a 2 mm distance is 2.67 μs . Thus, pulse widths in this range will provide the desired depth resolution. The lateral resolution (resolution along the beam diameter normal to the axis) is dominated by diffraction considerations.

The required *depth range* varies considerably for different anatomical studies. For example, a range of 25–30 cm is desirable for abdominal and obstetrical studies. For cardiac studies, the distance from the chest wall to the posterior heart wall is 15 cm or more. In superficial organs like the breast, thyroid, carotid and femoral arteries and in infant studies, the range of depth is 3–10 cm. Each of these depth ranges requires significantly different considerations as regards tissue absorption, specular reflection and small changes in tissue acoustic impedances. Generally high sensitivity receivers with wider dynamic range, low excitation frequencies and high transmitted powers help to extend the range. Focusing is usually necessary to achieve good resolution at the larger depths.

The *field of view* should be large enough to display the entire region under examination and to provide a useful perspective view of an organ of interest. When viewing small superficial structures such as the thyroid, a field of approximately 5 cm $\times$ 5 cm can encompass the desired region. For cardiac imaging, sector scans are preferable to rectilinear scans since a large structure is to be viewed through a small window. An angle of 60° is adequate to simultaneously view most of the heart. In abdominal studies, though the field of view is a matter of operator convenience, the viewed area should be at least 100 cm². It becomes difficult to view structures close to the surface when using sector scans because they fall in the apex of the sector. Therefore, two-dimensional arrays are used for abdominal studies.

In real-time imaging systems, the *frame rate* (the rate at which the image is repeated) should be rapid enough to resolve the important motions and to obtain the image without undesirable smearing. Most of these requirements are met with a frame rate of about 30 frames/s. This also satisfies the requirements for flicker-free display and is compatible with standard television formats. For some special studies, greater frame rates are often required for the data acquisition mode. These frames would then be stored and played back at about 30 frames/s to provide a slow motion presentation.

Detectivity is the ability of an imaging system to effectively capture, process and display the very wide dynamic range of signals which may, in turn, help to detect an image, lesion or other abnormal structure or process. Poor system detectivity manifests itself in lack of fidelity or picture quality which is often apparent in the visual displays of ultrasonic images.

23.7 MECHANICAL SECTOR SCANNER

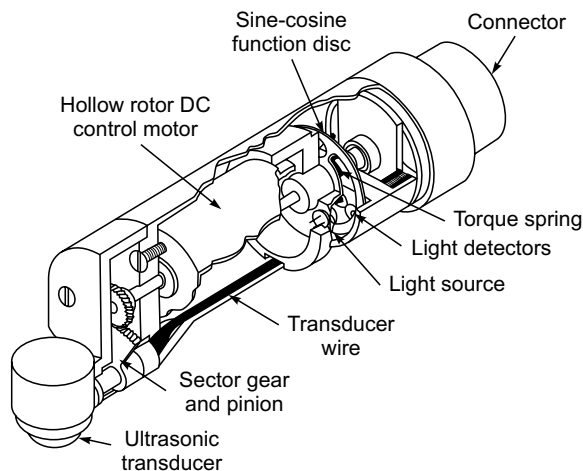
Mechanical scanning of a single transducer represents a low cost technique of extending the performance of a manually scanned system to achieve two-dimensional fields of view in real-time. In fact, the simplicity of the mechanical real-time sector scanner makes it an attractive device for medical applications. Several forms of motor-driven mechanical scanning of a single transducer have been used to achieve real-time two-dimensional imaging of the heart, the abdomen and the eye.

Schuetz *et al.* (1976) developed a sector scanning system (Fig. 23.17) which employs a single ultrasonic transducer (2.25 MHz) through an angle that is programmable from 0 to 25 degrees. The sector scan rate is variable from 0 to 40 scans per second and scan linearity is achieved by electronic programming of the transducer position to match that of a triangular waveform.

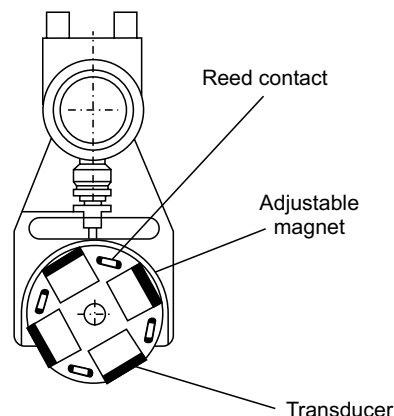
The transducer in the scanner is driven by a high performance, low inertia DC control motor. This motor incorporates a hollow-rotor armature yielding both low inertia and low inductance which are essential features for quick starting and stopping and for accurately following the rapid servo signals. The angular position of the transducer is determined photo-electrically from a sine-cosine function disc mounted on the rear of the motor shaft. Thus, the system incorporates many of the advantages of a small aperture electronically scanned array, while remaining compatible with most one-dimensional echocardiographic devices.

Holm *et al.* (1975) constructed a mechanical scanner consisting of a rotating wheel (Fig. 23.18) on which are mounted four identical unfocused 2 MHz transducers with a diameter of 2 cm. The transducers are placed radially at 90° intervals with their fronts at the periphery of the wheel. The wheel is driven, via a Bowden cable and chain by an asynchronous motor. The rotating scanner can be used hand-held or mounted on a conventional scanning arm. With 4 revolutions/s of the transducer wheel, 16 frames/s are displayed so that 50° sector images are produced in real-time. With each transducer sweeping through 90° in 1/16 s, the 50° sector scanned during the generation of 69.4 pulses giving 69 lines in each frame or 1.38 lines/degree.

By super-imposing several sector scans, a large-area trapezoidal scan can be obtained. This is achieved by interposing a real-time scanner and the compound scanner. Fig. 23.19 shows a sectional view of a mechanical

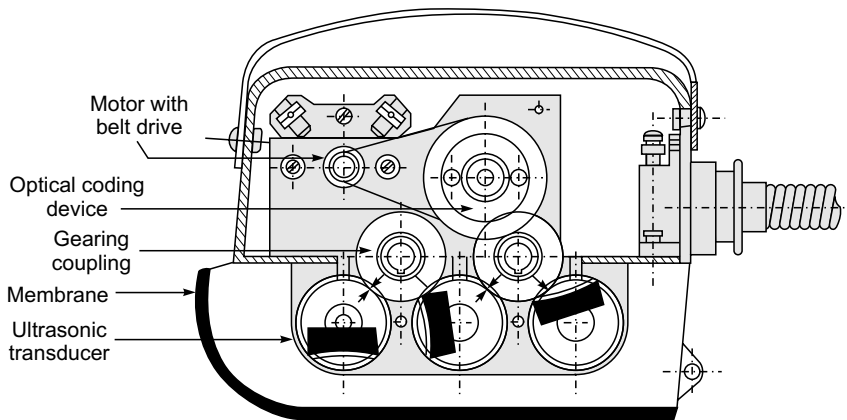


► Fig. 23.17 Details of mechanical scanner with a single transducer (After Schuetz *et al.*, 1976)



► Fig. 23.18 Construction of the rotating transducer type scanner (After Holm *et al.*, 1975)

trapezoidal scanning applicator. Three focusing individual probes rotate in a short intermediate path. Each probe covers a sector of 60° or 110° . The coupling membrane is shaped at the left side of the applicator in such a manner that it is also possible to take subcostal longitudinal views.



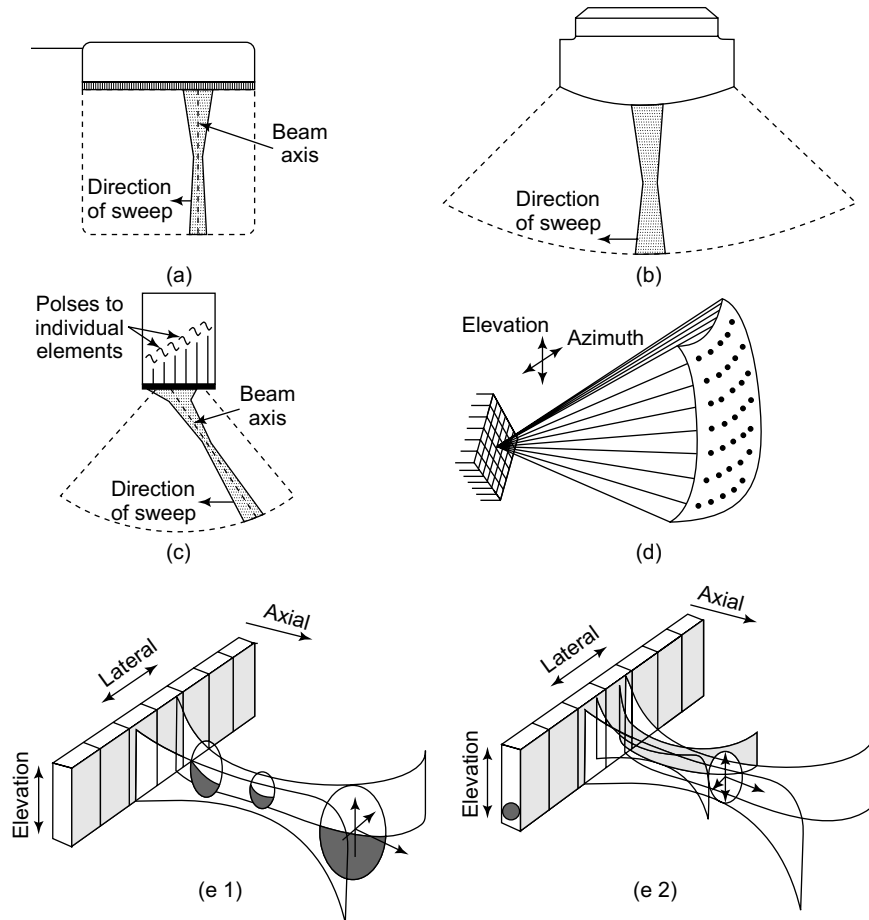
► Fig. 23.19 Three transducers mounted on three separate wheels for a larger field of view

23.8 MULTI-ELEMENT LINEAR ARRAY SCANNERS

Array transducers are available in several element configurations. Their arrangement determines the shape of the scan plane. For a two-dimensional image, the scanning plane is the azimuth dimension whereas the elevation dimension is perpendicular to the azimuth scanning plane. The transducers come in a variety of shapes, each containing a specified number of piezo-electric elements. The application usually determines the type and size of the transducer. There are three main types of transducer: linear, curved, and sector. Linear transducers are used primarily for small parts requiring high resolution and typically involve shallow depths. To achieve high resolution, higher frequencies are typically required. Since the elements are arranged in a linear fashion, the resulting image is rectangular. To broaden the field of view, sector transducers are used. These transducers have a small footprint and are good for cardio applications due to small rib spacing. For abdominal viewing, curved transducers are typically used because of resolution and penetration benefits. They allow for the maximum field of view and depth because of their large aperture. Note that any type of transducer can be phase arrayed to produce a beam of sound that can be steered and focused by the ultrasound controller. The shape of the region scanned for various array-element configurations are shown in Fig. 23.20.

Linear Sequential Arrays: The sequential linear array scans a rectangular region as shown in Fig. 23.20(a). The scanning lines are directed perpendicular to the face of the transducer. The beam is focused but not steered. Linear array transducers are available with 512 elements in currently available ultrasound scanners. Generally, up to 128 element are selected at a time for operation. As is obvious from the diagram, the field of view with a linear array arrangement is limited to the rectangular region directly in front of the transducer.

Curvi-linear Arrays: As shown in Fig. 23.20(b), a curvi-linear array scans a wider field of view because of the convex arrangement of the array elements. Curvi-linear or convex arrays operate in the same manner as the linear arrays and therefore, even in this case, scan lines are directed perpendicular to the transducer face.



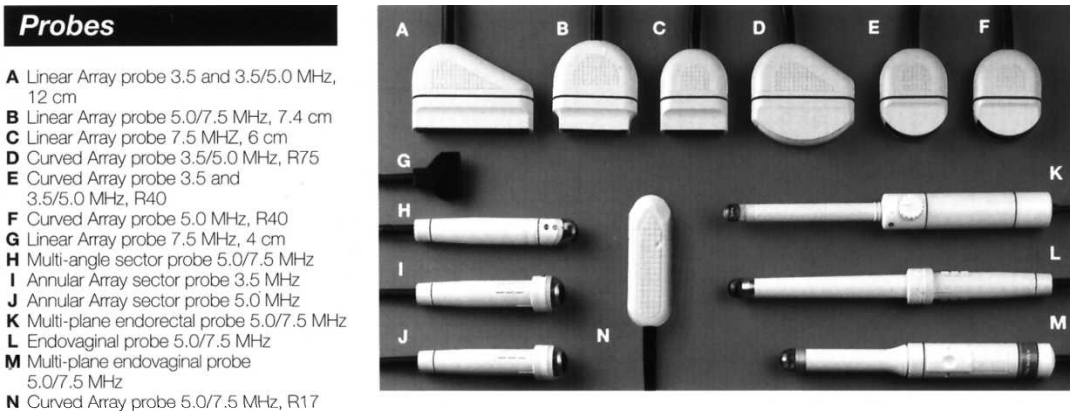
► **Fig. 23.20** Different arrangements in multi-element array scanners (a) linear sequential arrays (b) curvi-linear array (c) linear phased array (d) Phased array (e 1) Conventional Array Transducer: wider slice thickness in elevation direction with conventional array (e 2) Multi-D Array Transducer: Dynamic focusing in both the elevation direction and the lateral dimension provides narrower slice thickness and higher spatial resolution

Linear Phased Array: The arrangement of the elements is similar to the linear sequential arrays but the scanner steers the ultrasound beam through a sector shaped region in the azimuth plane. This is shown in Fig. 23.20(c). The present linear phased arrays systems have up to 128 elements and all the elements are used to transmit and receive each line of data. This arrangement has the advantage that the scanned region is much wider than the footprint of the transducer, thus making them suitable for scanning through restricted acoustic windows such as in the case of cardiac imaging wherein the transducer has to look through a small window to avoid the obstructions of the ribs and lungs.

Two-Dimensional Phased Array: A two-dimensional phased array has elements in both the azimuth and elevation planes. Therefore, two-dimensional arrays can focus and steer the acoustic beam in both dimensions, thereby to produce a volumetric image as shown in Fig. 23.20 (d).

Multi-focused transducers: The multi-focused transducers provide enhanced focusing capabilities by allocating more elements to the aperture than conventional system, with the effect of achieving exceptional spatial resolution throughout the depth of field [Fig. 23.20 (e)]. At the same time, these transducers allow ultra-fine control of the ultrasound beam pattern. The high degree of control over the beam pattern allows precise three-dimensional optimization of the beam resolution and suppresses side lobes at all depths, minimizing acoustic clutter for higher signal-to-noise ratio and improved spatial and contrast resolution.

The physical shape of the various types of ultrasound probes is shown in Fig. 23.21.



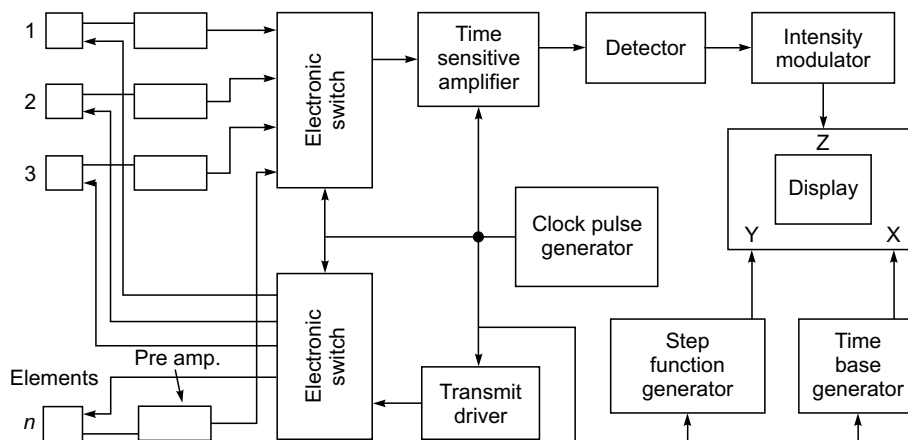
► Fig. 23.21 Various types of ultrasound probes used for real time scanning

23.8.1 Linear Array Scanners

The principle of linear array scanners is described by Bom *et al.* (1973) and can be summarized as the use in rapid succession of a number of parallel single elements with the display of each element in the brightness mode (B-mode) at almost the same instant. The transducer used comprises 20 elements placed on a line array of 8 cm. Each element measures 4 by 10 mm and the active area of the transducer is 80×10 mm. The depth penetration is designed for 16 cm, and therefore, the cross-section covered measures about 8×16 cm.

The elements transmit a short acoustic pulse sequentially into the tissue. Returning echoes are displayed along a horizontal axis on the CRT while the vertical position of each line corresponds to the position of the respective element. A block diagram of the principal components of the apparatus is given in Fig. 23.22. In transmission, a tone burst of the resonant frequency of the crystal is fed through an electronic switch into a single crystal element. The element generates a short acoustic pulse. The echo signals arriving at this element when in the reception mode are amplified by a wideband preamplifier and fed through a switch, into the processing part of the apparatus. This cycle is subsequently repeated for all elements. A time gain compensation circuit is used to compensate for the attenuation of ultrasound by spherical spreading and absorption in tissue. The display of the image can be on any of the standard devices.

In multi-element linear array transducers, the scanning action is based upon electronic beam stepping. However, the number of lines per scan is increased by using a group of elements as a composite transducer. This composite transducer may be progressively stepped, along the



► Fig. 23.22 Principle of multi-element scanning system. Block diagram shows major components in a multi-element signal processing and display system

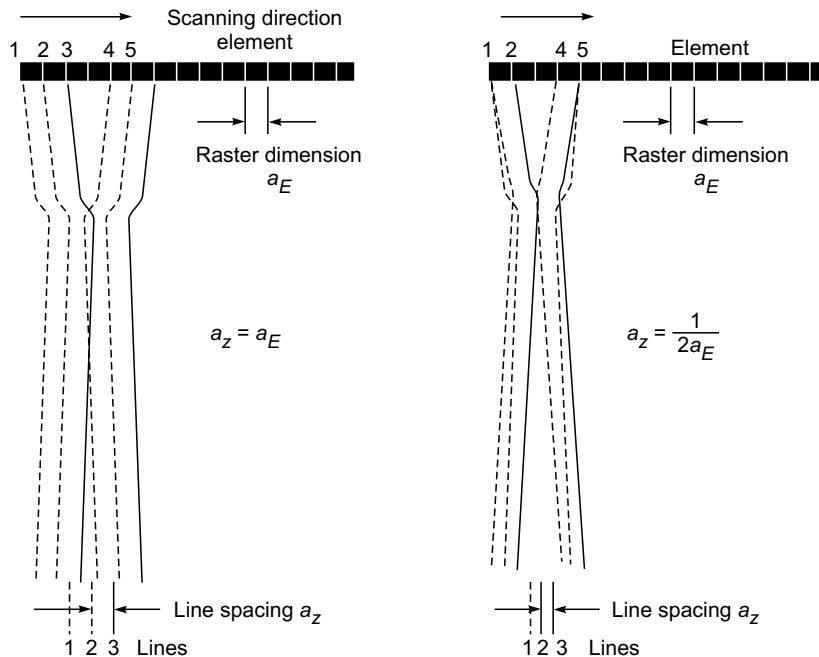
array in steps equal to the width of the narrow elements, by dropping the last element from the group and adding a new element to the front of the group. This produces a reasonably well-collimated beam to be moved through a series of overlapping positions along the array. Linear array applications currently available contain a large number of elements ranging from 64 to 420. They are usually activated in groups of eight, with progressive beam stepping in a sequence.

The linear array principle with beam stepping is described by Zurinski and Haerten (1978). The array applicators contain a linear array of 54 transducer elements. An initial pre-selected group of elements is actuated for the transmission of an ultrasonic pulse and receives the returning echoes.

These form one line of the two-dimensional image on the monitor. By switching in an element of the selected group on one side, the image line is shifted by one half an element width. In the next step, the image line is again shifted by another half an element width by switching off an element on the other side of the group. In this way, an image is built-up which consists of 101 lines. By selecting the group width, i.e., the number of elements per group, the position of optimum lateral resolution (focus) can be located in different depth ranges. With this system, the information content is doubled as compared with conventional techniques. This is achieved by a special type of stepped group-switching and does not involve line interlacing and multiple recording.

Normally, an element group is moved in the scanning direction in such a way that the first element of the group is switched off and at the same time, on the other side a new element is switched on (Fig. 23.23). The spacing between the individual scans corresponds to the element spacing (raster spacing).

The process is as follows: Firstly, a group consisting of m elements is selected ($m = 2, 3$ or 4). It transmits and receives in unison. A further element is then switched in without at the same time another one being switched off. This group has $m + 1$ elements. The central axis of the second group ($m + 1$) is shifted in the scanning direction by half the raster spacing in relation to the first group (m). After a new transmit receive cycle, the first element is switched off. The third group again has m elements and their centre axis is also shifted onwards by half a raster spacing. For the fourth group, a new element is switched in etc. The ultrasonic line sequence is



► Fig. 23.23 Group switching for selection of the focal distance in a multi-element scanning system (Redrawn after Zurinski and Haerten, 1978)

certainly recorded alternately with two different group widths and sound fields but this does not impair the information.

Delay lines are used to provide the required beam position and to shape in array transducers. To meet these requirements, the delay lines must have adequate bandwidth to preserve the pulse spectrum (2 MHz), large dynamic range (not less than 40 dB); tolerable insertion loss and adequate range (up to 10 microseconds), e.g., for 45° deflection in a 2 cm wide beam deflection array. Methods of providing electronically controlled delays include: (i) lumped constant delay lines with voltage-controlled capacitive elements; (ii) commutative delay lines where delay is determined by clock frequency; (iii) fixed delay lines, either tapped or combined in series by electronic switching; and (iv) charge coupled device (CCD) delay lines.

With rapid developments in technology, ultrasonic imaging systems are increasingly making use of digital techniques for beam forming and control, which provides better image quality than analog systems. In the case of the digital systems the received analog signal that comes from the transducer is digitized immediately. All the image processing and manipulation is done in the digital domain. In the case of the analog system, the digitization is done after a substantial amount of the processing has been done. The most important parameter in ultrasound image processing is the beam formation, which provides the focusing for the ultrasound beam.

In a transducer with multi-element crystals, the beam is focused and steered by exciting each of the elements at a different time so that the resulting ultrasound wave coming from each crystal will arrive at the intended focal point simultaneously. Incorporating delay lines usually does this. Obviously, digital time delays would permit much greater precision in the shaping of the ultrasound beam and do not require frequent calibration associated with analog beamformers.

Also, the digital beam former permits the system considerable flexibility in re-programming the size, shape, direction and intensity of the beam.

Although ultrasound resolution is affected by several factors, imaging frequency has the most direct influence. Ultrasound resolution improves in direct proportion to the frequency employed. In a typical 5-10 MHz system, the resolution cell measures roughly 0.7×0.35 mm. The result is that an anatomical structure smaller than 1 mm is likely to be missed.

Generally, the maximum imaging frequency is limited by the speed of the system's analog-to-digital converter. Conventional systems use A/D converters running at approximately 20 MHz. This limits the maximum imaging frequency to 10 MHz accordingly to the Nyquist sampling theorem. Moreover, as imaging frequencies increase, transducer design and fabrication become increasingly difficult. Further, the ultrasound penetration in the body decreases as the imaging frequency increases.

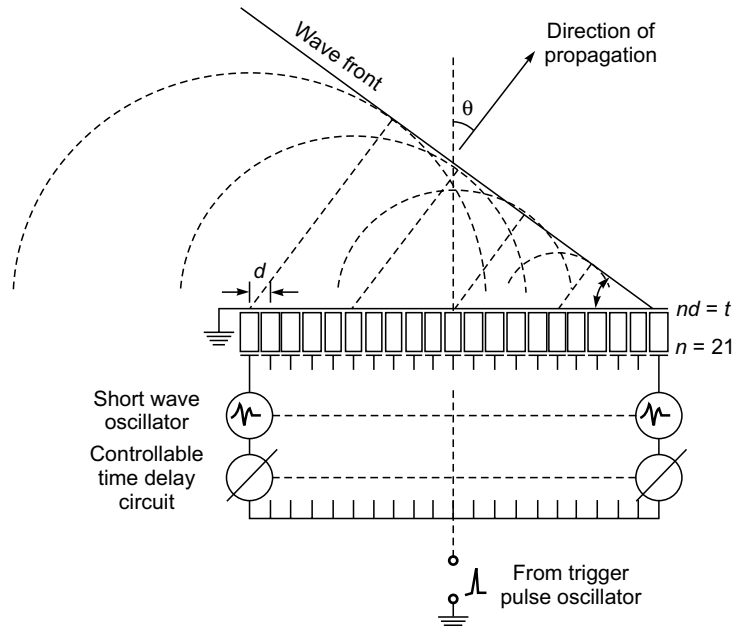
Recent advances in ultrasound transducer and electronics have enabled an extension of the ultrasound frequency for imaging beyond the conventional 5–10 MHz range. A high speed 40 MHz A/D converter has raised the Nyquist frequency, making 20 MHz imaging a possibility, to achieve a spatial resolution as fine as 70 micron. GE Medical offers a linear transducer which provides 12 MHz imaging frequency. With this system, objects smaller than 200 microns have been seen, opening an era of micron imaging. These systems have demonstrated axial, lateral and contrast resolution similar to that of Magnetic Resonance Imaging (MRI). Breast imaging is likely to see major benefits to achieve clearer visualization of the internal structure of lesions in the breast.

The linear array scanners mostly find applications in areas requiring a wide field of view such as obstetric scanning, rapid screening and orientation. On the other hand, real-time mechanical sector scanners are specifically designed for abdominal imaging for scanning organs such as the gall bladder, pancreas, and spleen, as well as hard-to-access areas of the liver and kidney and for structures such as those under the ribs or deep in the pelvis. Modern ultrasonic imaging systems sometimes incorporate both these techniques in single units in order to enhance their utility and economize on the cost by using common display for both the modes of operation.

Linear array transducers have increased versatility over mechanical transducers. Electronic scanning involves no moving parts and the focal point can be changed dynamically to any location in the scanning plane. The system is capable of generating a wide variety of scan formats. The disadvantages of linear arrays include the increased complexity and higher cost of the transducers. For high quality ultrasound images, many identical array elements are required. The array elements are typically less than a millimeter in thickness and each requires a separate connection to its own transmitter and receiver.

23.8.2 Phased Array System

Fig. 23.24 shows an electronic sector scanning technique which uses a multi-element array transducer. Each element is connected to its own independent oscillator, which can be triggered at the right moment. Triggering the local oscillators subsequently with equal time intervals will cause the individual wave-fronts to construct a resulting flat wave-front, the angle of which depends on these time intervals. If time delays of the wavefronts are $d \sin \theta / c$ (where c is the ultrasonic velocity and d is the inter-element distance) with respect to the reference element, then the direction of the main beam will have an angle θ with the normal. When these time intervals are zero, it means that all elements are excited simultaneously and the result is identical to that of a conventional transducer. Electronically variable pulse-delay circuits are used for delaying the trigger pulse which permits variation of the beam direction with great rapidity.



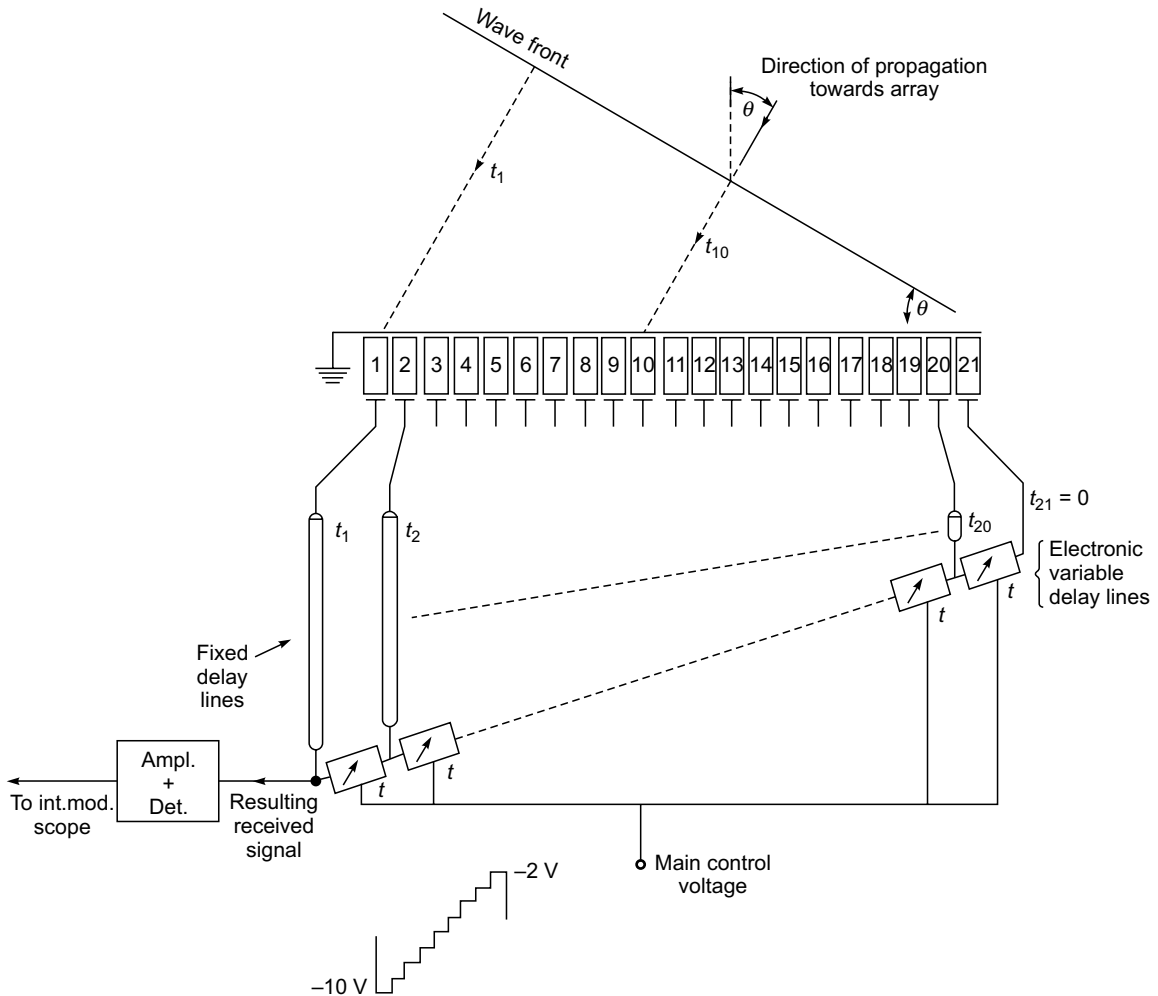
► Fig. 23.24 Beam pattern in a phased-array system

When using a pulse repetition rate of 1000 pulses per second, a sector can be scanned 30 times per second when 32 different directions are used. In practice, 128 lines are mostly used which means that every line on the screen is repeated about 8 times per second. A sector of 90° can be easily obtained using this technique.

For optimum lateral resolution, it is necessary to also make the reception of the echoes directional. Fig. 23.25 shows the principle of forming a received beam with an arbitrary direction using a system of fixed and electronically variable delay lines, which compensates for the differences in arrival time on the subsequent elements of an impinging wavefront from every direction. The same control voltage controls both the direction of transmission and that of reception, in order to always be maximally sensitive in the direction in which the pulses are transmitted. The control voltage also controls the direction of the line on the monitor screen. The techniques of real-time imaging with phased array ultrasound scanners are widely used in commercial scanners.

Phased array scanners are particularly useful for cardiac ultrasound sector scanning. A transducer measuring $1.2 \text{ cm} \times 1.3 \text{ cm}$ (2.25 MHz) or $1.2 \text{ cm} \times 0.8 \text{ cm}$ (3.5 MHz) is commonly used. The transducer is thus small enough to be easily manipulated by hand and the beam is small enough at the transducer face to be directed through the inter-costal spaces in any desired orientation. It yields an 84° sector angle, and produces 64 shades of gray to all sector displays. The gray scale controls allow an emphasis on select echo amplitude ranges. For example, it is possible to visualize such structures better as valve orifices during studies of mitral or aortic stenosis.

A typical phased-array ultrasound system will have from 32 to as many as 256 transmitters and receivers. In many cases, the system will have fewer transmitters and receivers than the number of available transducer elements. In these cases, high-voltage switches located in the transducer or system are used as multiplexers to connect a specific transducer element to a specific transmitter/receiver (Tx/Rx) pair. In this way, the system can dynamically change



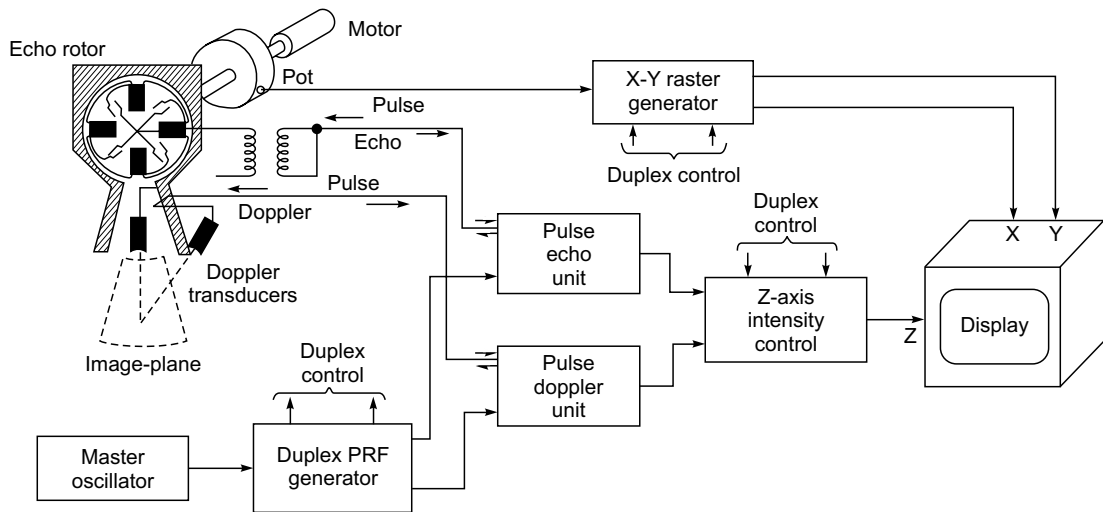
► Fig. 23.25 Principle of forming the received beam in a phased array system

the active transducer aperture over the available transducer element array. The requirements for these switches are severe. They must handle transmit pulses with voltage swings as large as 200VP-P and with peak currents up to 2A. They must switch rapidly to quickly modify the configuration of the active aperture and maximize image frame rate.

23.9 DUPLEX SCANNER

Duplex echo-Doppler scanning has been found to be useful for direct non-invasive measurement of peripheral arterial flow and tissue geometry. Here, the echo image is used to locate the artery and determine the position of the walls. The echo image is then used as a guide to map out the flow field with the Doppler. The incorporation of separate echo and Doppler transducers in a duplex scanner results in high resolution images due to the fact that the echo transducer beam can be aligned perpendicular to the artery wall, while the flow probe is at an angle appropriate for detecting Doppler shift.

Fig. 23.26 shows the major components of the duplex scanner (Barbaro and Macellavi, 1978). It basically comprises the scanner head, the pulse-echo unit, the pulse Doppler unit, the duplex control unit and the display.



► Fig. 23.26 Block diagram of Duplex scanner (After Barber et al., 1974)

Echo Rotor: Pulse-echo imaging is performed by rotating a transducer so that its beam sweeps out the image plane. The rotor contains four identical transducers but at any one time, only the one pointing in the direction of the image plane is used. The proper transducer is automatically selected by an external magnet and the reed switches shown in the rotor diagram. The proper transducer is connected through its switch to a rotary transformer to the pulse-echo unit. The rotor is driven by a DC motor and the position of the rotor is measured with a precision sine-cosine potentiometer. The operation of the system, in most respects, is therefore, independent of the speed of the motor.

The echo transducers have a 5 MHz piezo-ceramic element that is 1 cm in diameter. Each element is focused with an epoxy lens with a focal length of about 5 cm.

Doppler Transducer: There are two Doppler transducers, selectable by a slide switch mounted in the wall of the housing. With the switch in the position shown, the pulse-Doppler is connected to the transducer whose beam is contained within the image plane, and is fixed at an angle of 37.5° to the vertical. The other transducer is not in the plane of the image and is shown in projection on the diagram. Its beam intersects the image plane at a single point at an angle of 37.5° to the plane. This transducer is useful when one wishes to visualize an artery in cross-section, in which case the flow would be perpendicular to the image plane and the Doppler shift measured with the other Doppler transducer would be theoretically zero.

The pulse-echo unit contains a pulser, a 5 MHz broadband amplifier with time-compensated gain, and a video detector. The pulse-Doppler unit also runs at 5 MHz. The output going to the Z-axis intensity control is normally a pulse taken from the range-gate generator. Most ultrasound systems provide two to as many as four switchable transducer connectors to allow the clinician to easily switch among the various transducers for each examination.

The system runs off a 5 MHz master oscillator. The echo period is the time between each pulse from the echo PRF pulse train and is 100 μ s. The Doppler period is 60 μ s, a time which is

shorter because the Doppler transducer is located near the skin surface. These and other logic signals control the echo and Doppler units, and the display circuitry.

The rotor potentiometer, a ramp generator and the duplex control signals are combined to generate both echo and Doppler x-y signals. The intensity of the display is quantized to four discrete shades of gray. The echo video signal is converted to a set of parallel binary signals which are re-combined with the Doppler range information, determined by the duplex control. This real-time quantization technique greatly enhances the apparent resolution of the B-mode image.

Modern duplex scanner systems are microcomputer-based which provide dual frequency pulsed Doppler with two-dimensional imaging, A-mode and gray scale M-mode. The dual frequency Doppler operates at a frequency of 3 MHz for abdominal and adult cardiology studies and at 5 MHz for peripheral vascular disease and for carotid and paediatric flow studies. Just as echocardiography has greatly expanded our ability to understand spatial anatomic relationships of structures within the heart, the ability to simultaneously depict blood flow in a cross-sectional plane would greatly enhance understanding of the physiology of blood flow in the normal and diseased heart. In recent years, devices have become commercially available which literally allow one to see the blood flow through the heart along with echocardiography. The technique uses moving target indicator procedures to detect reflections from blood cells as they move through the scanned cross-section. The direction of flow is displayed in colour with one colour to display flow towards the transducers and another to display flow away from the transducer. The velocity of flow is estimated from the display by the brightness of the colour display.

In *continuous wave Doppler* (CWD), the ultrasound transducer elements are split into equal halves about the centre of the transducer aperture. Half of the elements are used as transmitters to produce a focused acoustic CWD transmit beam; the other half of the elements serve as receivers to produce a focused receive beam. The signals applied to the transmit elements are square waves at the Doppler frequency of interest, typically 1 MHz to 7.5 MHz. Transmit jitter needs to be minimized to avoid phase-noise generation that can adversely affect Doppler phase-shift detection. The transmit beam is focused by properly phasing the signals applied to the transmit elements. In a similar way, the CWD received signals are focused by phasing and summing the signals from each receive element. Because the transmitters are on simultaneously with the receivers in this mode, the Doppler signals of interest are typically within a few kilohertz of a very large receive signal that is generated by reflections from stationary tissue at the transmit fundamental.

In *spectral Doppler* processing, the beam-formed digital signals are digitally filtered, mixed to baseband by using quadrature local oscillators (LOs) at the transmit frequency, and then sampled at the transmit pulse repetition frequency (PRF). A complex, fast Fourier transform (FFT) is used to generate an output spectrum representing the velocity content of the signal. The signal magnitude for each bin of the FFT output is calculated and compressed to optimize the available, visible display dynamic range. The signal magnitude is finally displayed versus time on the ultrasound display. The spectral processor also generates left and right stereo audio signals that represent positive and negative velocities. A DAC converts these signals which are used to drive external speakers and headphones.

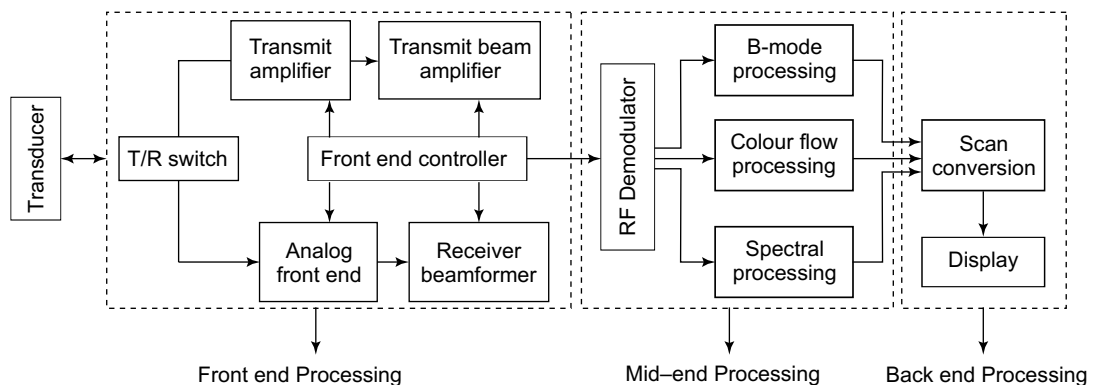
In *colour-flow* processing, the RF digital beam formed data is digitally mixed by using quadrature local oscillators (LOs) at the transmit frequency to do the complex mixing into baseband signals. As a result, each sample of the acoustic receive line has associated magnitude and phase values assigned. In colour-flow processing, 8 to 16 acoustic lines are typically collected along the same image path line in order to measure Doppler shifts. Reflections from moving blood flow or from

moving tissue along that image path will create a Doppler shift and, therefore, change the phase of the baseband samples where that shift occurred. The colour processor determines the average phase shift versus time for each point along that image path over the 8 to 16 lines; the processor also assigns a colour to represent that average velocity. In this way, a two-dimensional colour representation of blood or tissue motion is made.

23.10 MODERN ULTRASOUND IMAGING SYSTEMS

Modern ultrasound imaging systems make use of digital signal processors and system-on-chips which are specially designed single-chip microcomputers that process digitized electrical signals generated by transducers that have revolutionized the area of diagnostic ultrasound imaging. A diagnostic ultrasound imaging system generates and transmits acoustic waves and captures reflections that are then transformed into visual images. The signal processing on the received acoustic waves include interpolation, decimation, data filtering and reconstruction. Programmable DSPs and SOC (System-on-Chip), with architectures designed for implementing complex mathematical algorithms in real-time, can efficiently address all the processing needs of such a system (Pailoor and Pradhan, 2008).

The basic functional building blocks of a diagnostic ultrasound imaging system are: the transducer, processing blocks needed for image formation, system controller and power supply module as shown in Fig. 23.27. Delineation of signal path operations as front-end, mid-end and back-end processing varies from one manufacturer to another. It also depends, to some degree, on the type of technology used for carrying out these operations: application-specific integrated circuit (ASIC), field-programmable gate array (FPGA), DSP or PC.



► Fig. 23.27 Ultrasound imaging system functional blocks

The transmit beamformer is responsible for the pulse-excitation of transducer elements, which results in emission of acoustic waves into the region of interest. Immediately following this, the transmit/receive (T/R) switch is positioned to put the front end in receive mode. Transducer elements transform the reflections or echoes into corresponding electrical signals. The analog front end (AFE) properly amplifies these signals and converts them into digital data streams for further processing. By applying dynamic delays into these data streams, the receive beamformer combines them to form a scan line, a representation of the region of interest along a given line of sight. This process is repeated either sequentially or simultaneously to form multiple scan lines to cover a region of interest. The front-end controller is responsible for controlling the timing

and sequencing of transmit and receive beams. Sampling rates used for analog-to-digital (A/D) conversion in the front end can vary from 16 MHz to 50 MHz depending on system requirements.

Most ultrasound imaging systems are capable of three major modes of operation: B-mode, colour-flow and Doppler. B-mode operation results in a gray scale image that is used for examining tissue structures and organs. Colour-flow operation results in a colour-coded display of spatial distribution of mean velocity of blood flow super-imposed on gray scale image. Doppler processing produces scrolling display of blood flow velocity distribution at a user specified location. Some systems are capable of displaying all three modes simultaneously. Common to all three is the initial stage where beamformed RF data gets down converted to baseband, filtered, and in most cases, decimated. What follows this depends on the mode. For B-mode, it is envelope detection and logarithmic compression. What needs to take place for colour-flow is a lot more compute intensive processing involving high pass filtering of ensembles of scan lines to remove contributions from vessel wall or tissue motion. Output of the wall filter is then used to estimate power, mean velocity, and turbulence. Doppler processing involves a much simpler filter and estimation of velocity distribution using short-time Fourier transform techniques. Doppler processing also produces a stereo audio signal representing the Doppler spectrum.

Hereafter, B-mode and colour-flow estimates are subjected to temporal and spatial processing to reduce noise and enhance features of interest. Scan conversion is the operation where B-mode and colour-flow estimates are converted to display raster data, pixels with 1:1 aspect ratio. When colour-flow is on, B-mode and colour-flow pixels have to be blended to produce a single image. This blending is typically based on application dependent thresholds.

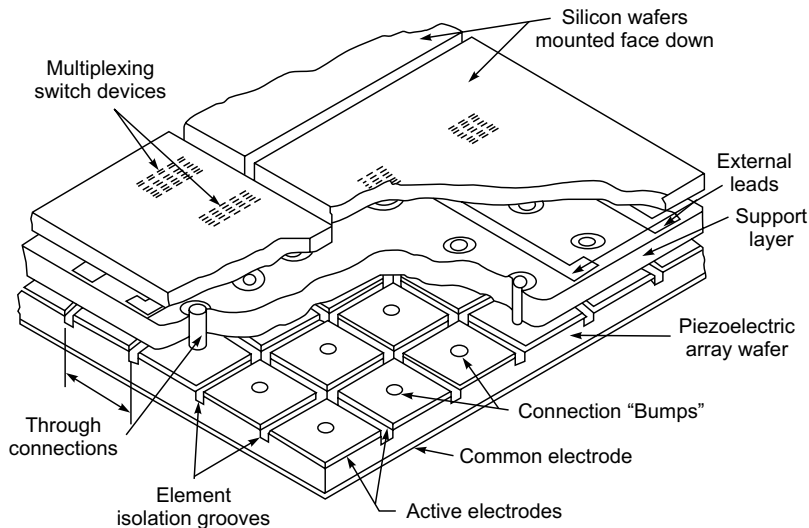
Just like most embedded systems, ultrasound imaging systems need a system controller to carry out functions such as configure and control the signal path, events and take appropriate actions, acoustic pressure and intensity levels and ensure safety of patients. A typical ultrasound scanner is shown in use in Fig. 23.28.



► Fig. 23.28 *Ultrasound scanner in use*

23.11 AREA ARRAY SYSTEMS

For achieving high spatial resolution throughout a volumetric field of view, two-dimensional area arrays of transducers have been developed. In this system, an image of any place within the object can be focused on a two-dimensional transmit-receive array of piezo-electric transducers by means of an acoustic lens. The two-dimensional area array (Fig. 23.29) consists of 16×16 matrix of 2.25×2.25 mm transducer elements. A module of 16×16 matrices is used to form a 32×32 element array. The operating frequencies may be 2.25 or 3.5 MHz. In the time sequence, each element in the array operates in a pulse-echo mode as both the transmitter and receiver of ultrasonic energy. Accessing individual elements within this array is possible through an x-y address system consisting of row and column inter-connect lines and double-diffused metal-oxide-semiconductor transistor (DMOST) multiplex switches, co-located with each transducer element. Applying address signals to a single x-control line and a single y-control line connects the transmitter and receiver to one



► **Fig. 23.29** Construction of two-dimensional ultrasonic array using integrated circuit technique (After Meindl, 1976)

and only one transducer element. The piezo-electric elements and multiplexer switches are made in an integrated form. The acoustic lens provides high lateral resolution, a large depth of focus and a sector scan pattern which requires a minimal body aperture.

The area-array transducer permits operations in several different modes (Meindl, 1976). Any full row or column of elements may be scanned using either intensity modulation (B-scan) or beam deflection modulation (multiple A-scan) to produce cross-sectional displays. Electron switching from row-to-row or column-to-column at a slow rate provides the capability for online three-dimensional target visualization throughout a pyramidal volume. The range and frame rate of the system can be controlled to obtain deep body imaging in real-time. Thus accurate dynamic measurements of heart chamber volumes are quite feasible in this system. However, the area array involves a rather complex electronic circuitry considering the number of transmitters, receivers, delay lines and other circuits, etc.

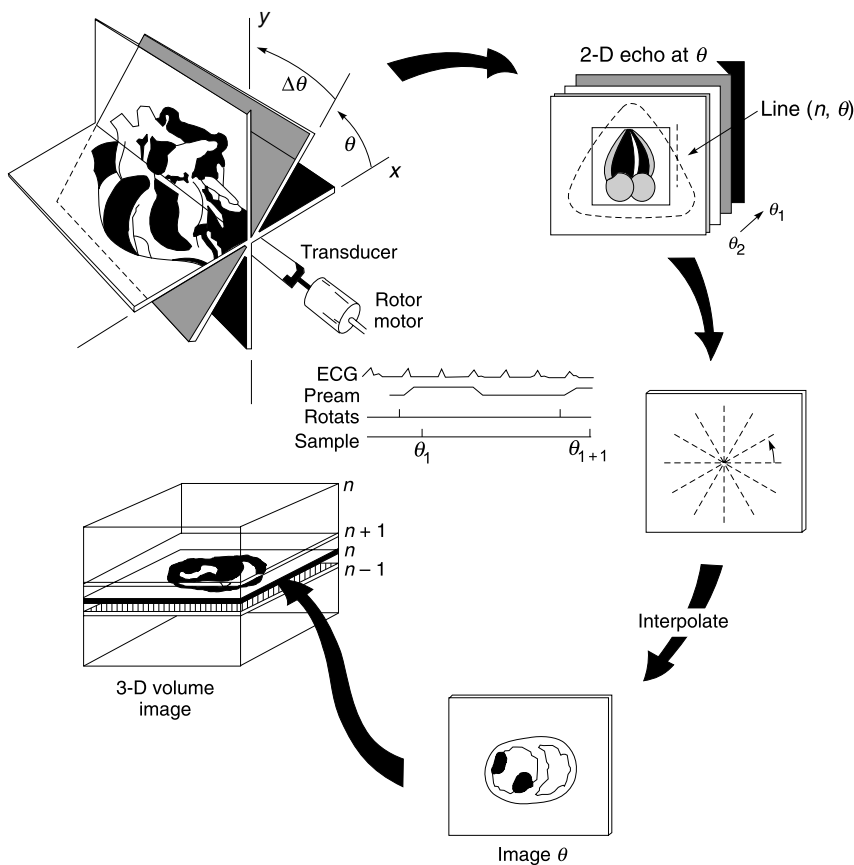
23.12 THREE-DIMENSIONAL ULTRASOUND IMAGING SYSTEMS

Ultrasonic imaging portrays qualitative information (for example, shape and physical context, and tissue type) and quantitative information (for example, organ dimensions, tissue properties such as attenuation and speed). Two-dimensional echocardiography allows comprehensive exploration of the structure and function of the heart by mental integration of the information from multiple cross-sectional views. Accordingly, a computer based three-dimensional reconstruction is a logical step towards improved appreciation of complex cardiac structures and reliable quantification of volumes.

Extensive research work in the field of three-dimensional echo-cardiography has been going on since the introduction of cardiac ultrasound imaging was established as a diagnostic tool. Its clinical implementation has not become widely popular because of a number of technical problems related to data acquisition and reconstruction. Specifically, these problems include spatial location of the image planes, temporal alignment of the image frames within the cardiac

cycle, correction for respiratory motion of the heart, rapid data transfer, storage of large amounts of data, development of suitable algorithms for reconstruction and finally, display of three-dimensional images on a two-dimensional screen.

The most widely applied approach (Roelandt *et al.*, 1994) for the three-dimensional image acquisition is the rotation of the scan plane around a central axis from a fixed transducer position. For data acquisition, the image plane is rotated in regular increments spanning a total angle of 180° . In this way a cone-shaped volume is scanned with its apex located at the transducer position. For each scanning position, sequential images covering an entire cardiac cycle are required and transferred either to the hard disk of the workstation or to the optical disk of the ultrasound system. Images can be stored using a video or a digital format. In order to obtain homogeneous data sets, only image sequences from cycles that fall within pre-selected limits of the R-R interval, are accepted. The final data set consists of the gray level values of all pixels with their exact spatial location and their temporal location within the cardiac cycle. For processing and reconstruction, data is later retrieved by disk by independent three-dimensional workstation. For three-dimensional display, the reconstructed volume can be dissected along any selected plane and cardiac structures of interest can be viewed from different angles. Several algorithms such as distance coding are used to create the illusion of depth. Data can be displayed



► Fig. 23.30 Principle of three-dimensional imaging system (redrawn after McConn *et al.*, 1988)

as a dynamic loop sequence of an entire cardiac cycle with moving structures viewed from a selected angle. McCann *et al.* (1988) illustrates the *in vivo* and *in vitro* techniques for three-dimensional ultrasound imaging for cardiology.

The principle of three-dimensional imaging, in a simplified form, is illustrated in Fig. 23.30 (McCann *et al.*, 1988). The transducer (5 MHz) is attached to a stepper motor which rotates through 180° to obtain 100 views at 1.8° increments. One level (n) from each of the section scans is stacked into the volume image at the proper angle to generate the n th section of the image. The whole sequence is recorded. A sequence of frames is selected and digitized to generate a $512 \times 512 \times 8$ bit image for each angle of view. By the process of sequential stacking of the sector scans at various positions of the transducer, the final volume image is constructed. Comprehensive user friendly software packages are now used in commercial scanners to give a three-dimensional display of scanned images.

The 3-D imaging technique is being popularly utilized for viewing the foetus during various stages of pregnancy, wherein highly exciting pictures of the unborn child can be stored as an album. Fig. 23.31 shows 3-D image of a baby at 33 weeks of pregnancy. A logical extension of the 3-D images is the 4-D ultrasound image which brings to the user an image which is continuously updated and it becomes a moving image like a movie.



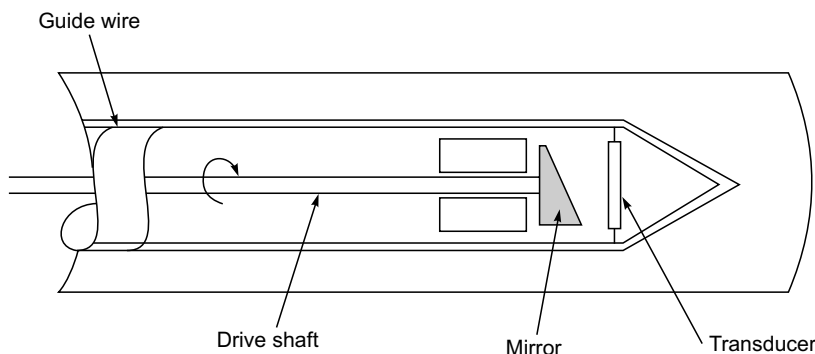
► Fig. 23.31 A 3D image of baby at 33 weeks picking its nose (Courtesy: www.fetalvisionimaging.com)

23.13 INTRAVASCULAR IMAGING

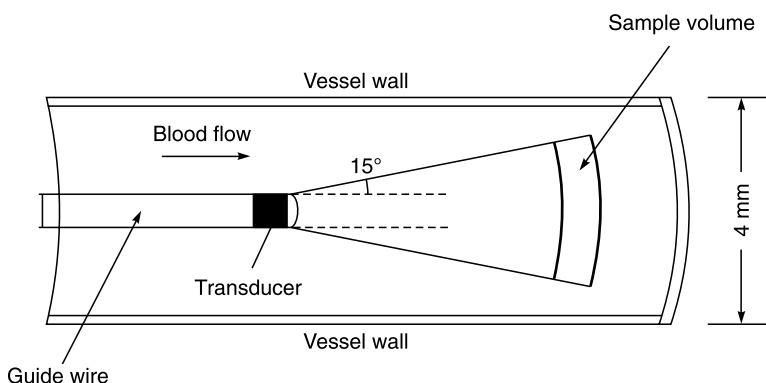
Intravascular ultrasound has recently attracted considerable interest because it yields high-resolution images not only of the vessel lumen but has the unique potential of visualizing the vessel wall and pathology. Although intravascular imaging has a long history of research behind it, beginning in the early 1970s with the pioneering work by Bom *et al.* (1972), it is only recently that it has shown potential for intra-cardiac application.

The normal first order coronary is 2–5 mm in diameter, but second order vessels and atherosclerotic lumina often reach 0.1–2.0 mm. The small size of these vessels puts a limit on the size of the transducer and the acoustic power to obtain a favourable signal-to-noise ratio. The transducer is normally carried on the tip of a catheter, which should be flexible enough for safe passage into the vessels. The flexibility not only ensures that the catheter can be manoeuvred to a central and coaxial position in the vessel but also enhances both safety and image quality. Various technical approaches have been employed to construct an ultrasound probe capable of intra-luminal examination (Nissen and Gurley, 1991). Basically, there are two fundamental approaches: one utilizing a mechanically rotated transducer or acoustic mirror, the other utilizing an electronic array.

Fig. 23.32 shows an intra-luminal ultrasound catheter probe for two-dimensional imaging of the coronary arteries. The device comprises a fixed transducer element at the tip of the catheter and a rotating mirror driven by an external motor and a flexible drift shaft. By using this device, we can obtain radial cross-sectional images of the artery. After placing a guide wire in the coronary artery, the catheter is passed by guide wire control into the selected area of the artery and positioned under fluoroscopic control. The outer diameter of the catheter is typically 1.4



► Fig. 23.32 Schematic diagram of intraluminal ultrasound catheter device



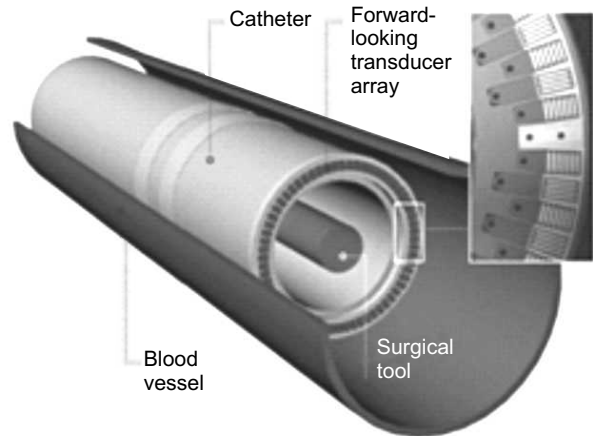
► Fig. 23.33 Schematic diagram of intraluminal doppler guide wire device

mm. At an ultrasound frequency of 30 MHz, high resolution images of the vascular pathology are obtained. Despite their apparent simplicity, mechanical transducers are difficult and costly to manufacture within consistent mechanical tolerances (Bommer and Haerten, 1991).

In another approach, very small Doppler probes have been incorporated into guide wire tips for introduction into the coronary artery. Fig. 23.33 shows a schematic illustration of the Doppler guide wire. A piezo-electric element is incorporated into the tip of a 0.014" or 0.018" guide wire to produce a forward looking pulsed Doppler signal at typically 15 MHz.

The multi-element electronic array uses 1.83 mm diameter, 20 MHz intravascular ultrasound probe with no mechanical parts. The device employs an array of 64 transducer elements mounted at the tip of a 4.5 F catheter. The design permits the central lumen to accommodate a standard 0.014" guide wire to facilitate placement in the artery. The electronic signals are amplified and multiplexed by micro-miniature integrated circuits within the transducer assembly. The image reconstruction algorithm is highly complex and utilizes a principle known as a 'synthetic aperture array' and not the conventional phased array approach. With the synthetic aperture array, the multi-element transducer sends and receives ultrasound signals sequentially from each of the elements. Multiple elements contribute to the reconstruction of more than 1000 radial scan lines at 10 frames per second. For nearby pixels, only a few adjacent elements are utilized, whereas more distant targets employ larger groups of elements. This variable aperture approach provides a device that theoretically remains focused from the surface of the catheter to infinity.

MEMS fabrication techniques have enabled to construct a capacitive micromachined ultrasonic transducer, referred to as CMUT or silicon ultrasound. These transducers can be easily formed into arrays of various shapes and sizes. Small ring-shaped arrays, for example, can provide forward-looking ultrasonic views for doctors probing an artery with a catheter (Fig. 23.34). Dangerous blockages and certain kinds of heart problems can then be treated using surgical tools that pass through the centre of the catheter. The elements of the array shown in this diagram are each composed of many individual transducer units. The capacitive transducers are made by using photolithography and other fabrication techniques of the semiconductor industry and the transducer arrays—large or small and even with the most complex geometries can be made very precisely and inexpensively.



► **Fig. 23.34** *Micromachined capacitive ultrasound transducer*

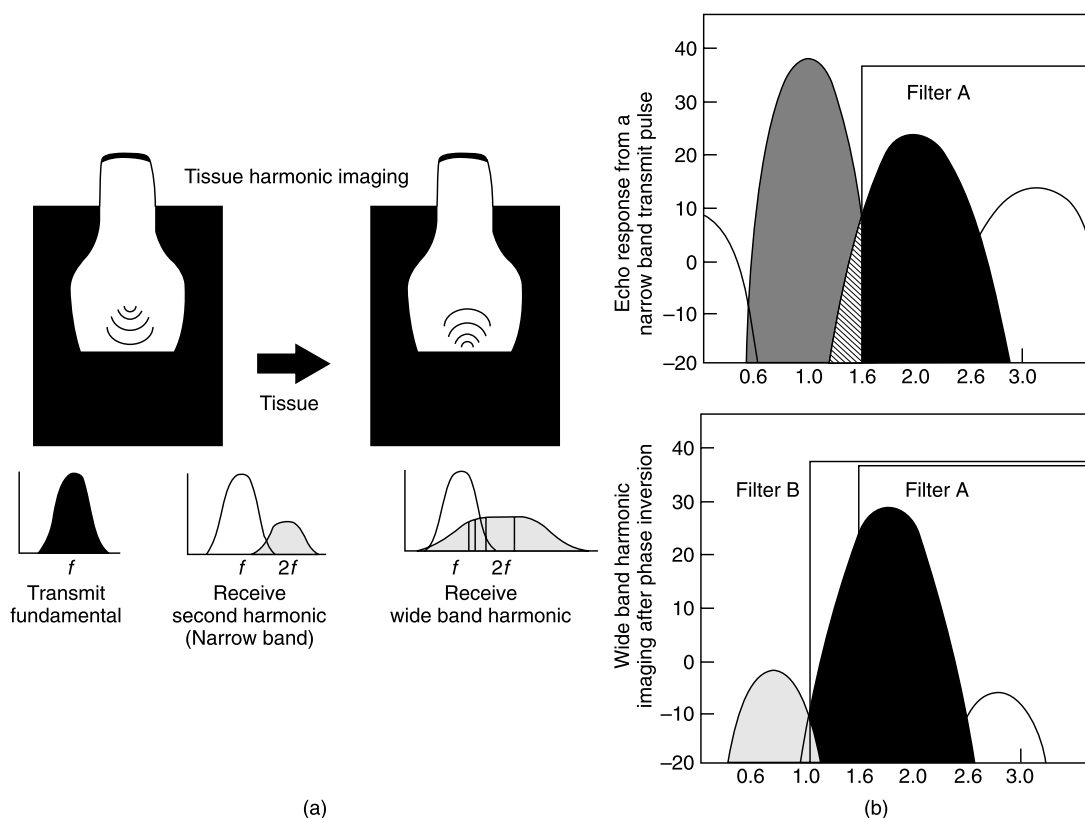
The capacitive transducers have greater bandwidth than piezo-electrics. The difference arises because a piezoelectric transducer is by nature a highly tuned device, like the pendulum of a clock. At its particular resonant frequency, a piezoelectric transducer undergoes high-amplitude oscillations, even with very little forcing, but at other frequencies, it barely moves at all—which is to say that it has very limited bandwidth. A capacitive transducer also has a distinct resonant frequency, but only when it's operating in air. When it's immersed in water or coupled to biological tissues, which are much like water in their acoustic properties—the situation becomes very different. Because the vibrating membrane has so little mass, its movements become highly damped by the watery medium it touches. This effect, then, lets a single transducer work well over a broad swath of the ultrasonic spectrum. It means that the transducer is able to emit and detect the many different frequencies that are contained in a short ultrasonic pulse. The shorter the pulse you use to probe the patient's body, of course, the better the depth resolution in the resulting image.

■■■■► 23.14 TISSUE HARMONIC IMAGING

When an ultrasound signal pulse propagates through the target media, a fundamental change occurs in the shape and consequently frequency of the transmitted signal. A sound wave propagating through the body is, in fact, a pressure wave. This pressure wave compresses and relaxes the tissue. Where tissue is compressed, the speed of sound is higher, effectively causing the top of the pressure wave to be pulled forward. As the tissue relaxes, the speed of sound becomes lower. The distortion of the waveform causes harmonics to be generated. For example, a transmitted frequency of 3.0 MHz will also return a harmonic frequency of 6.0 MHz (second harmonic) and 9.0 MHz (third harmonic). In view of the availability of harmonics in the returned echo signals combined with the advanced signal processing techniques, a new ultrasonic imaging technology called tissue harmonic imaging has recently been developed (Haerten *et al.*, 1999). Reception of these second-harmonic signals has, for a variety of reasons, produced better image quality and is now widely used.

In tissue harmonic imaging systems, the returning high frequency signal has to travel in only one direction i.e. return to the probe. The displayed image would have the benefits from the attributes of high frequency imaging and a one-way travel effect. Apart from increase in axial, lateral and contrast resolution, there is a substantial decrease in beam aberration, reverberation and side lobes. Harmonic imaging is most efficient in the mid-to far-range before absorption limits its penetration.

The conventional method of generating ultrasonic images based on harmonics uses a narrow band pass filter to separate the strong fundamental echo components from the harmonic components. This results in a loss of contrast resolution due to the inherent overlap of the fundamental echo with the harmonic signal, which in turn, results in both reduced spatial resolution and contrast resolution. Using phase inversion technique in which two consecutive pulses are transmitted into the body, with the phase of the second inverted relative to the phase of the first, the echoes from the linear body structure get cancelled whereas echoes from non-linear structures do not. This



► **Fig. 23.35** Fundamental imaging : Transmit frequency f = Receive frequency a) Middle: 2nd Harmonic imaging-transmit frequency f , receive frequency $2f$. b) Right : Wide band harmonic imaging – transmit frequency f , receive all frequencies. (a) Echo frequency response from a narrow band transmit pulse. (i) using a high pass filter or narrow band filter (filter A) cannot effectively separate the fundamental from the harmonic signal due to the overlap of the signals. (ii) After phase inversion, the summing of the even harmonic components results in greater harmonic signal amplitude. Wider band pass filtering can now be applied more effectively (filter B). The result is improved contrast and spatial resolution

results in higher amplification of even harmonic components giving superior contrast and spatial resolution. A wide band pass filter is used to receive all echo signals. The impact of this technique on the imaging performance is shown in Fig. 23.35 which compares the relative lateral beam profiles of the fundamental and the harmonic signals. Side lobes which are responsible for artifacts affecting the imaging quality are significantly reduced as compared to conventional imaging. Images appear sharper and crisper with less noise and with higher contrast resolution.

23.15 PORTABLE ULTRASOUND SYSTEMS

Over the past several years, ultrasound equipment has become more compact, with cart-based systems increasingly complemented or replaced by portable and handheld ultrasound machines. This reduction in size is enabling a wide variety of health care applications that illustrate how advances in medical technology are bringing care to patients instead of forcing them to travel. Portable and handheld ultrasound systems are also instrumental in bringing health care to rural and remote areas, disaster sites, patient rooms in hospitals, assisted-living facilities and even ambulances.

Single-chip ultrasounds systems are currently under development to enable portable designs, but has historically also required a trade-off in image quality, which limited the unit's usefulness as a diagnostic tool. However, recent advances in integration have minimized that trade-off by reducing the signal path's size, as well as noise levels. The efforts are to create a single-chip design that allows the probe, receive, transmit and processing chains to be integrated. That in turn allows all of the ultrasound machine electronics to reside in the probe head, which then can use a wireless link to transmit information to the display. Single-chip designs also complement 3-D probes, which have significantly more transducer elements and thus require more cabling to the machine.

23.15.1 Mobile Phone-based Ultrasound Imaging System

Smart phones are being increasingly used as bio-medical devices as a profusion of new software applications or apps are becoming available which will play a larger role in healthcare. This has become possible as smart phones have powerful processors, large memory, an ample touch screen with amazing graphics and built-in wireless connectivity for e-mail and internet access. They have enough power to run third party apps and peripherals making them a valuable platform for mobile applications (Mertz, 2012). Fig. 23.36 shows a complete package that includes a cell phone and an ultrasonic transducer that plugs into the phone. The system comes with the app-installed phone, with a universal serial bus (USB) host port, which most mobile phones lack. Ordinary phones have a USB port but it is for charging. They do not have host port to support reading a flash drive. The nurses or community health workers in remote areas can be easily trained to take good ultrasound images and for interpretation they can transmit the image to a more highly trained radiologist. The system offers both cost and portability advantage.

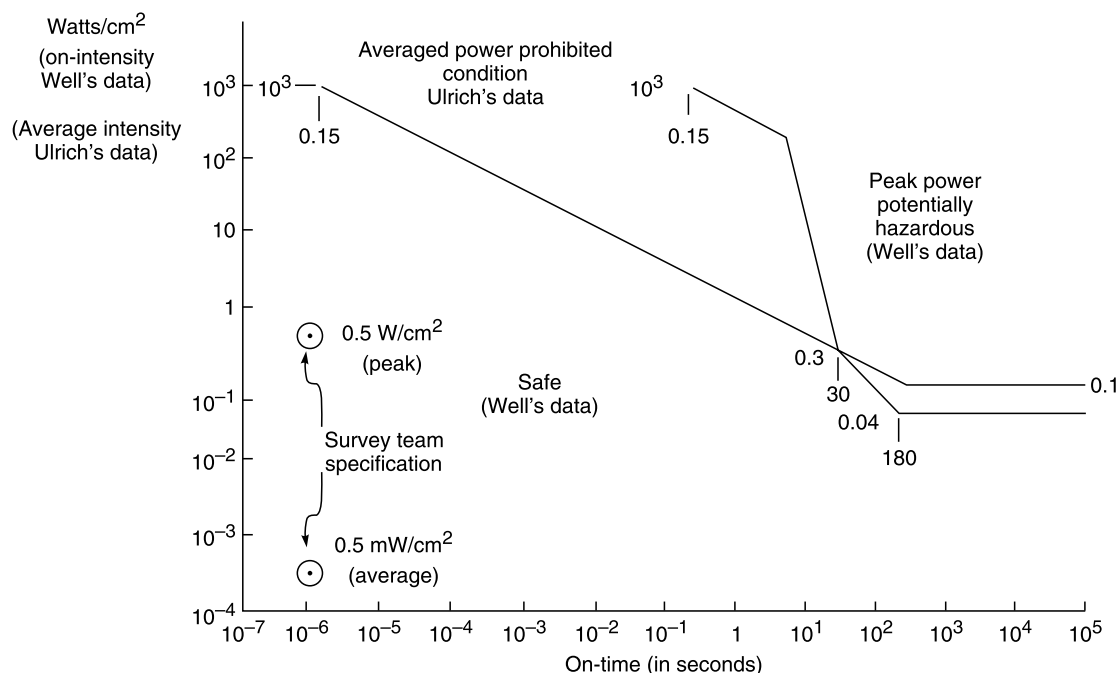


➤ Fig. 23.36 A portable mobile phone and ultrasound scanning system (Courtesy: Mobisante, U.S.A.)

23.16 BIOLOGICAL EFFECTS OF ULTRASOUND

Ultrasound methods are attractive for diagnosis because of their non-invasive nature and their apparent safety. Ultrasonic radiation in clinical use is known to be substantially less hazardous to tissues than radiations like X-rays which are in common use in some of the same applications. The apparent safety of ultrasound rests on data from experimental animal work and over 40 years of extensive human clinical experience. The experiments conducted include studies on whole animals, individual organs, isolated cultured cells and biochemical studies. Surveys of human populations which have been exposed to ultrasound, though on a limited scale, have also been taken. The types of damage sought include genetic and teratogenic effects, disorders of physiological function and inhibition of biological development (Hussey, 1975). The consensus of all these experimental results has supported the current medical opinions that no significant hazard is associated with the levels of ultrasound used in diagnosis.

Ulrich (1971) prepared a schedule of safe ultrasound dosage for experimental use on human beings (Fig. 23.37). He observed that the zone of safety for continuous wave ultrasound lay below a log/log line connecting 1 ms of 1 kW/cm² ultrasound, with 200 s of 100 mW/cm² ultrasound. An ultrasound intensity of 100 mW/cm² or less was also safe for at least 10,000 s. Pulsed wave ultrasound was safe for doses in which the average intensity multiplied by the total exposure time (or peak intensity multiplied by sum of pulse on-time) lay within this zone of safety. The zone of safety was valid for 0.5 to 15 MHz and for all anatomic sites except the eyes. Re-exposures were limited to 10% and 30% per year. The data in this figure provide the best estimates which can be made on the basis of the available evidence of 'safe' conditions



► Fig. 23.37 'Safezone' in ultrasonic applications at different power levels for continuous and pulse-mode technique (Courtesy: National Science Foundation, U.S.A.)

of ultrasonic diagnosis. Presently, it is recommended that ultrasonic diagnosis should not be permitted unless the irradiation falls within the safe conditions defined in Fig. 23.36. On the basis of Wells (1977) data, a pulse-echo system operating with pulses of 1 ms duration and 10 W/cm^2 on intensity, at a pulse repetition frequency of 100/s should not be used to examine any individual patient for a total time exceeding 2.75h. On the other hand, a Doppler detector (continuous wave) operating at an intensity of less than 0.04 Wcm^{-2} may be used without time restriction.

An exhaustive treatment of the subject on safety and potential hazards in the current application of ultrasound in general, and in obstetrics and gynaecology in particular has been rendered by Lele (1979). Experiments have failed to reveal any evidence that tissues have a 'memory' for exposure to ultrasound. Thus, unlike X-rays and other ionizing radiations, there is a little likelihood of cumulative effects of ultrasound. Also, ultrasound, because of its long wavelengths in tissues, is unlikely to cause intra-cellular focal damage such as that resulting from X-radiation.

The necessity for ultrasound safety also captured the attention of the American College of Obstetrics and Gynecology (ACOG), as evidenced by their Committee Opinion # 180 entitled "New Ultrasound Output Display Standard" issued in November 1996. This publication described the onscreen display standards (as determined by the FDA, the AIUM, ACOG, and the National Electrical Manufacturers Association – NEMA) for commercially available ultrasound systems. The ultrasound devices manufactured after 1992, one of two acoustic output indices (MI or TI) is required to appear on the screen.

Thermal index (TI): is a calculated estimate of temperature increase with tissue absorption of ultrasound and is determined by the ratio of the total acoustic power to the acoustic power required to raise the tissue temperature by 1°C .

Mechanical index (MI): The mechanical effects of ultrasound absorption are also known as non-thermal effects and are represented by the mechanical index (MI), which is a relative measure. The MI is calculated by dividing the spatial-peak value of the peak rarefactional pressure (rated by 0.3 dB/cm-MHz at each point along the beam axis) by the square root of the centre frequency. Specifically, the mechanical effects are the result of compression and decompression of insonated tissue with the formation of microbubbles, also referred to as cavitation.

Current standards recommend that if an ultrasound device is capable of achieving a TI or MI greater than 1.0, then the output display screen must show the appropriate index value for the user to predict the potential for adverse bioeffects. The specific recommendation for obstetrical ultrasonography is for the MI or TI not to exceed 1.0.

Finally, prudent use of ultrasound is recommended with the ALARA principle that mandates that ultrasound exposure (in terms of TI, MI, and exposure time) be kept "as low as reasonably achievable" without compromising diagnostic capability or resolution. The American Institute of Ultrasound in Medicine (AIUM) in March 1993 issued the following official statement on clinical safety. "Although the possibility exists that such biological effects may be identified in the future, current data indicate that the benefits to the patient of the prudent use of diagnostic ultrasound outweigh the risks, if any, that may be present."

■■■■► **MODEL QUESTIONS** ◀■■■■

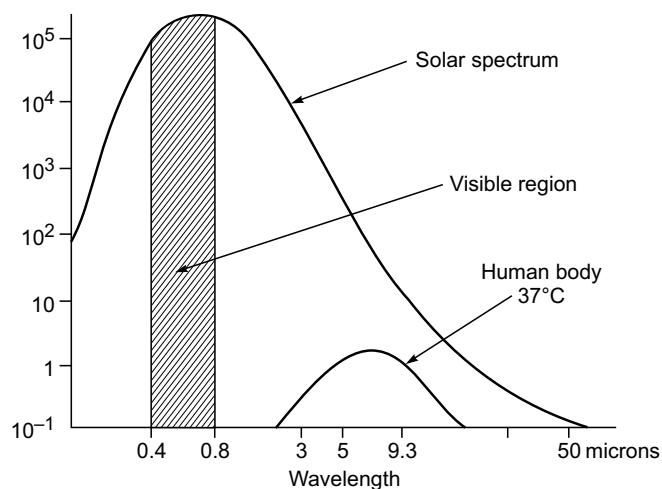
1. What are the advantages of an ultrasound imaging system?
2. Define characteristic impedance in ultrasound. Why a coupling medium is essential when using ultrasound for diagnostic purposes? What is the frequency range of ultrasound normally used in medical field for diagnostic applications?
3. Define 'Resolution' with respect to ultrasound systems. What are the two types of resolution in an ultrasound system? Explain them.
4. How is ultrasound generated? Explain the role of frequency, focussing and active element diameter with reference to ultrasound transducers.
5. Describe with the help of a diagram the various building blocks of a basic pulse-echo apparatus.
6. Explain the construction of an ultrasound probe used for pulse-echo systems. Why multilayer matching is necessary in the probe?
7. Distinguish between 'A' and 'B' scan displays with the help of a diagram. What are the common applications of these two types of scan modes?
8. What is 'M' mode display in an ultrasound system? Name the instrument used in medical practice based on M-mode display. Describe its working with the help of a block diagram.
9. What is a real time ultrasound imaging system? What are special requirements of real time imaging systems?
10. Explain the constructional and working details of the following types of real time transducers:
 - (a) Mechanical sector scanner
 - (b) Multi-element linear array scanner
 - (c) Linear array scanners
 - (d) Phased array scanner
 - (e) Area array scanner
 - (f) Duplex scanner
11. Describe the working principle of linear array scanner with the help of a block diagram.
12. Explain with the help of a diagram the working of ultrasound tissue harmonic imaging system.
13. Explain the principle of constructing 3-D ultrasound images.

Thermal Imaging Systems

24.1 MEDICAL THERMOGRAPHY

The medical thermograph is a sensitive infrared camera which presents a video image of the temperature distribution over the surface of the skin. This image enables temperature differences to be seen instantaneously, providing fairly good evidence of any abnormality. However, thermography still cannot be considered as a diagnostic technique comparable to radiography. Radiography provides essential information on anatomical structures and abnormalities while thermography indicates metabolic process and circulation changes, so the two techniques are complementary.

The human body absorbs infrared radiation almost without reflection, and at the same time, emits part of its own thermal energy in the form of infrared radiation. The intensity of this radiant energy corresponds to the temperature of the radiant surface. It is, therefore, possible to measure the varying intensity of radiation at a certain distance from the body and thus determine the surface temperature. Fig. 24.1 shows spectral distribution of Infrared emission from human skin.



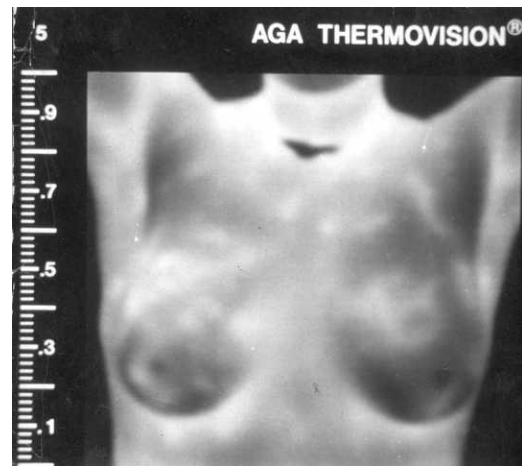
► Fig. 24.1 Spectral distribution of infrared emission from human skin. The emission peaks at around 9 microns regardless of pigmentation

In a normal healthy subject, the body temperature may vary considerably from time to time, but the skin temperature pattern generally demonstrates characteristic features, and a remarkably consistent bilateral symmetry. Thermography is the science of visualizing these patterns and determining any deviations from the normal brought about by pathological changes.

Thermography often facilitates detection of pathological changes before any other method of investigation, and in some circumstances, is the only diagnostic aid available. Thermography has a number of distinct advantages over other imaging systems. It is completely non-invasive, there is no contact between the patient and system as with ultrasonography, and there is no radiation hazard as with X-rays. In addition, thermography is a real-time system.

The examination of the female breast as a reliable aid for diagnosing breast cancer is probably the best known application of thermography. The mammary glands were the first organs that thermography was clinically applied (Lawson, 1957) to. It is assumed that since cancer tissue metabolizes more actively than other tissues and thus has a higher temperature, the heat produced is conveyed to the skin surface resulting in a higher temperature in the skin directly over the malignancy than in other regions. Fig. 24.2 shows typical breast thermograms.

Stark and Way (1974) have been carrying out a study of women, who, in the absence of any breast pathology, have abnormal or asymmetrical thermograms. They found that the incidence of subsequent development of breast cancer in these women is upto 10 times greater than those whose thermograms have been symmetrical; therefore, thermographic examination is of considerable value in identifying a risk population. Thermography also finds applications in the assessment and monitoring of inflammatory joint diseases, diagnosing deep vein thrombosis (Arter, 1978) and the problem of peripheral circulation. Anbar (1998) covers recent developments in clinical thermography in many fields of medicine from general surgery to ophthalmology.



► Fig. 24.2 Normal gray-tone thermogram of female breast

24.2 PHYSICS OF THERMOGRAPHY

24.2.1 Infrared Radiation

The infrared ray is a kind of electromagnetic wave with a frequency higher than the radio frequencies and lower than visible light frequencies. The Infrared region of the electromagnetic spectrum is usually taken as 0.77 and 100 μm . For convenience, it is often split into near infrared (0.77 to 1.5 μm), middle infrared (1.5 to 6 μm) and far infrared (6 to 40 μm) and far infrared (40 to 100 μm). Infrared rays are radiated spontaneously by all objects having a temperature above absolute zero. The total energy 'W' emitted by the object and its temperature are related by the Stefan-Boltzman formula,

$$W = \sigma \epsilon T^4 \quad (i)$$

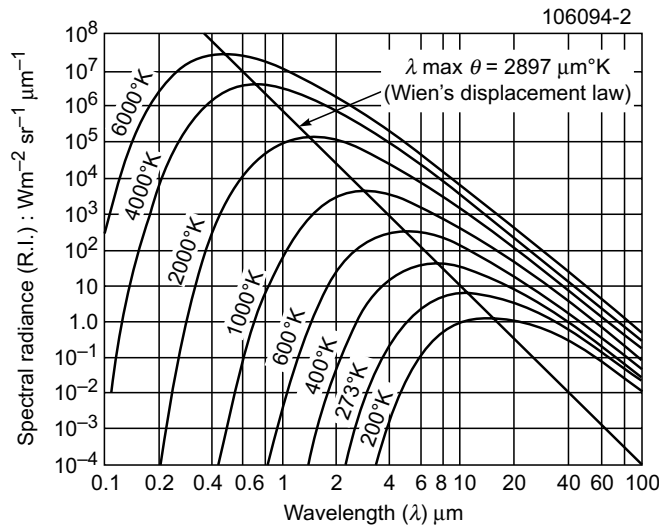
where W = radiant flux density and is expressed in W/cm^2

ϵ = Emissivity factor

σ = Stefan-Boltzman constant = $5.67 \times 10^{-12} \text{ W}/(\text{cm}^2 \times \text{K}^4)$

T = Absolute temperature

Equation (i) shows that the amount of infrared energy emitted varies with temperature of the object. Fig. 24.3 shows the spectral distribution of intensity of radiation from black bodies. The wavelength of the energy peak and the absolute temperature is given by the Wien formula,



► Fig. 24.3 Black body spectral radiance

$$\lambda_{\max} = \frac{2897 (\mu\text{m})}{T(\text{K})}$$

The human body has a temperature of 37°C (310 K), therefore,

$$\lambda_{\max}(\text{human body}) \approx 10 \mu\text{m}$$

From equation (i), Fig. 24.2, the energy density of the infrared radiation of the human body = $4.6 \times 10^{-2} \text{ W}/\text{cm}^2$. Assuming the surface area of the human body to be $1.5\text{--}2.0 \text{ m}^2$, the amount of infrared energy radiating from the whole body is approximately 700–1000 W.

24.2.2 Physical Factors

There are several physical factors which affect the amount of infrared radiation from the human body. These factors are emissivity, reflectivity and transmittance or absorption.

Emissivity: An object which absorbs all radiation incident upon it, at all wavelengths, is called a black body. A black body is only an idealized case and, therefore, all objects encountered in practice can be termed gray bodies. We thus define the term emissivity as representing the ratio of the radiant energy emitted per unit area by an object to the radiant energy emitted per unit area of the black body at the same temperature.

$$\epsilon = \frac{W_o}{W_b}$$

The value of ϵ for human skin at ambient temperature is virtually unity (Watmough and Oliver, 1968) within the limit of infrared wavelengths of 3–16 μm . If a body has an emissivity that is less than one, and constant whatever the wavelength, it is called a gray body. A change in emissivity causes an error in temperature measurement, Sakurai *et al.*, (1973) calculated that 1% of $\Delta\epsilon$ is equivalent to approximately $0.3^\circ\text{C } \Delta T$.

The spectral radiant emissivity is defined as

$$\epsilon_\lambda = \frac{W_{o\lambda}}{W_{b\lambda}}$$

where $W_{o\lambda}$ is the spectral radiant power per unit area and $W_{b\lambda}$ is the same quantity for a black body. ϵ_λ is a function of both wavelength and temperature, but the variation with temperature is usually much smaller than the variation with wavelength.

In the infrared region, the spectral radiant emissivity of most solids decreases as the wavelength increases.

Reflection: Spectral reflectivity ρ_λ is defined as the ratio of reflected power to the incident power at a given wavelength. So,

$$\rho_\lambda + \alpha_\lambda = 1 \quad (i)$$

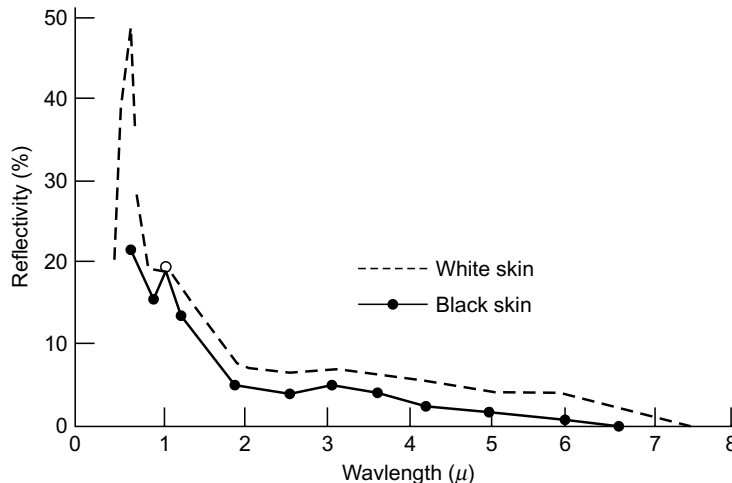
where α_λ is the spectral absorptance. Since

$$\epsilon_\lambda = \alpha_\lambda$$

where ϵ_λ is the spectral emissivity, then

$$\epsilon_\lambda = 1 - \rho_\lambda \quad (ii)$$

Therefore, an opaque body having a low emissivity in visible light will have high reflectivity. Hardy (1939) illustrated the spectral distribution of the reflectivity ρ_λ of human skin. This is shown in Fig. 24.4. It shows that the reflectivity of human skin between the range 2 and 6 μm is 0.02–0.05. Obviously, a thermogram will be affected by strong infrared light coming from



► Fig. 24.4 Spectral distribution of the reflectivity of human skin (After Hardy, 1939)

sources outside the body and getting reflected from the body. Incandescent lights emit infrared radiation and can influence a thermogram. Equation (ii) shows that if emissivity decreases, there is a corresponding increase in the reflection coefficient.

Transmittance and Absorption of Infrared Radiation: When a semi-transparent body is placed between the surface of any radiation-emitting body and a detector, it is necessary to consider the change in emissivity related to its transmittance, reflectivity and emissivity. The presence of tape, paste or ointments has influence on thermograms.

Hardy (1939) studied the spectral absorption of infrared radiance through human skin and found that the human skin is almost opaque at the wavelengths encountered in thermography.

The background radiation of infrared rays is also found to influence the thermograms. When the infrared radiation from the background and that of the body are equal or nearly equal, it is difficult to distinguish the body from the background.

24.3 INFRARED DETECTORS

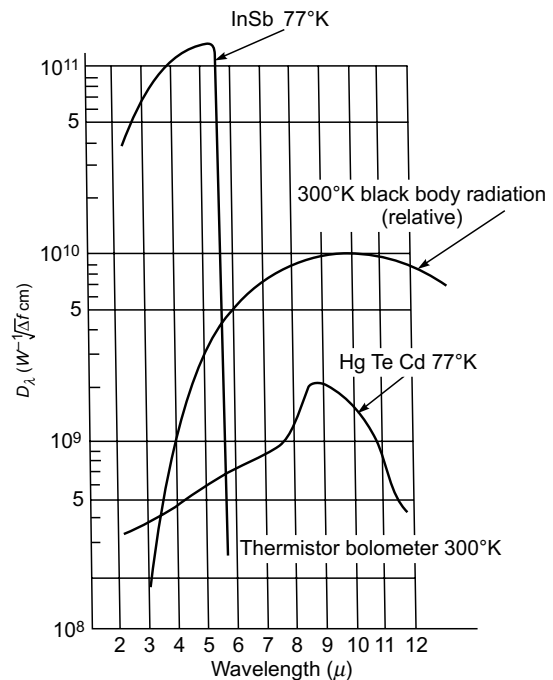
Infrared detectors are used to convert infrared energy into electrical signals. Basically, there are two types of detectors: thermal detectors and photo-detectors.

Thermal detectors include thermocouples and thermistor bolometers. They feature constant sensitivity over a long wavelength region. However, they are characterized by long-time constant, and thus show a slow response.

The wavelength at which the human body has maximum response is 9–10 μm . Therefore, the detector should ideally have a constant spectral sensitivity in the 3–20 μm infrared range.

However, the spectral response of the photo-detectors is highly limited. Most of the infrared cameras use InSb (indium antimonide) detector which detects infrared rays in the range 2–6 μm . Only 2.4% of the energy emitted by the human body falls within the region detected by InSb detectors. But they are highly sensitive and are capable of detecting small temperature variations as compared to a thermistor.

Another detector making use of an alloy of cadmium, mercury and telluride (CMT) and cooled with liquid nitrogen, has a peak response at 10–12 μm . The spectral response of various detectors used in thermographic equipment is shown in Fig. 24.5.



► Fig. 24.5 Spectral characteristics of various types of infrared detectors

24.4 THERMOGRAPHIC EQUIPMENT

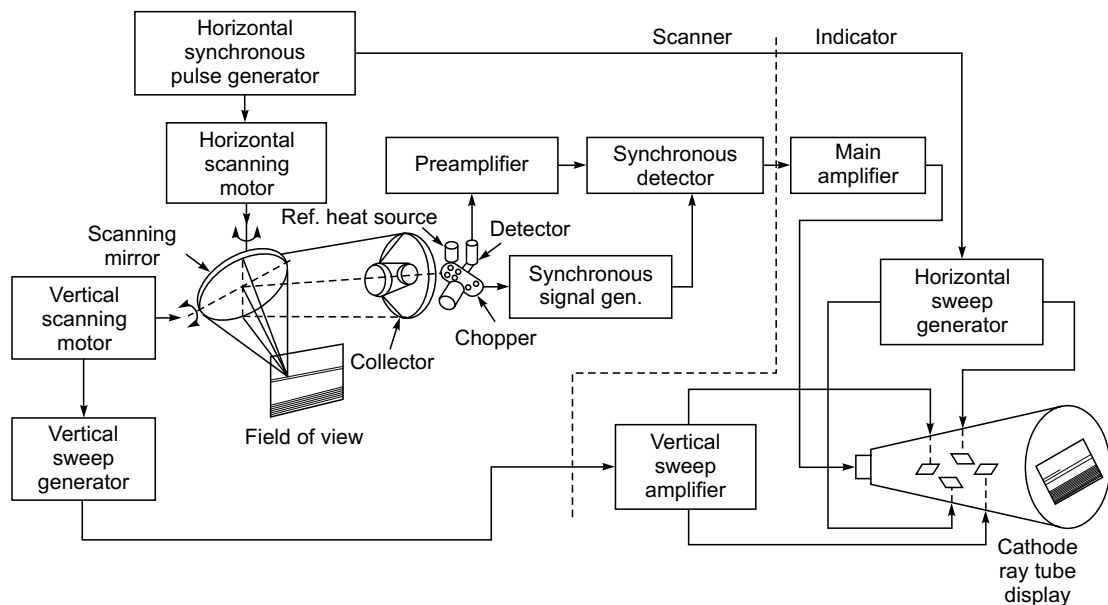
Thermographic cameras incorporate scanning systems which enable the infrared radiation emitted from the surface of the skin within the field of view to be focused on to an infrared detector. Most systems have a wide range of absolute temperature sensitivity ranging from 1 to

50°C. However, for most of the examinations, the gray scale is generally adjusted to represent a much narrower range of temperature, depending on the area to be examined. A full scale (black to white) temperature difference of between 5 to 10°C is usually adequate. For breast examinations, the absolute temperature range may be 25 to 35°C, for the legs 23 to 33°C and for the forehead 31 to 36°C (Carter, 1978). With the gray-scale spanning a temperature range of 5°C, it is possible to resolve temperature variations of 0.5°C on the skin. Most of the clinical changes in temperature are of the order of 1°C or more. It is not necessary to use the system on a more sensitive mode, even though some systems can theoretically resolve 0.1°C.

The equipment used in thermography basically consists of two units: a special infrared camera that scans the object, and a display unit for displaying the thermal picture on the screen. The camera is generally mounted on a tripod that is fitted on wheels.

The camera unit contains an optical system which scans the field of view at a very high speed and focuses the infrared radiation on a detector that converts the radiation signal into an electrical signal. The signal from the camera is amplified and processed before being used to modulate the intensity of the beam in the picture tube. The beam sweeps across the tube face in a pattern corresponding to the scanning pattern of the camera. The picture on the screen can be adjusted for contrast (temperature range) and brightness (temperature level) by means of controls on the display unit.

Fig. 24.6 shows the schematic arrangement of an infrared scanner. The double-scanning movement of the plane mirror causes each spot on the patient's body to be focused in turn on the cooled detector of Indium Antimonide. The detector is mounted in a Dewar flask which is cooled with nitrogen in order to obtain optimum sensitivity and eliminate thermal noise. An optical collector system is employed in an infrared detecting system. This is due to a weak infrared input from the subject. Scanning is carried out by a reflecting mirror. The instantaneous field of view which can be scanned is decided by the optics and detector size.



► Fig. 24.6 Block diagram of the scanning and displaying arrangements for infrared imaging

The horizontal and vertical movements of the scanning mirror are controlled by individual motors. The scanning mechanism employs an arrangement for rapidly oscillating the flat mirror to give a horizontal line scan, and slowly tilting it for the frame scan. Simultaneously, horizontal synchronous pulses and vertical sweep signals for the display unit are also generated. A chopper disc interrupts the infrared beam so that an AC signal is produced from the detector. The chopping disc also interrupts light from a standard heat source, which intermittently illuminates a photo-cell and provides a phase-reference to a coherent detector and to a bucking circuit employed to cancel out large standing signals. The adjustment of the height and the tilt of the optical head can be controlled to have a different field of view.

The AC signal developed at the detector is amplified by the pre-amplifier. It is then rectified and fed into the bandpass filter whose central frequency is determined by the chopping frequency. The total voltage gain in the amplifier is around 120 dB. A CRT is used for display of scanned image. Horizontal sweep, which is synchronized with the scanning motion of the scanning mirror, is displayed from top to bottom and then in the reverse order on the screen. The signals are displayed on the screen by intensity modulation which controls brightness and contrast with the strength of the signal. Each picture is built from around 150 horizontal lines in about 75 s. Temperature differences of the order of 0.1°C can be resolved.

In the commercial instruments based on this technique, the object is scanned in a television-like raster wherein each line consists of temperature information divided into picture elements. The thermal picture on the screen is virtually flicker-free as a result of the high frame rate of 16 frames per second. In spite of the high frame rate, temperature differences of less than 0.2°C can be detected. The resolving power is 140 standard lines per picture and roughly 130 elements per line. The size of the picture on the display tube is 90×90 mm.

The system is equipped with a $10^{\circ} \times 10^{\circ}$ field of view front lens providing a minimum object size of 11.5×11.5 cm at a distance of 0.95 cm. By an additional extension ring, it is possible to cover an object as small as 4.6×4.6 cm.

The thermal and spatial resolution of a thermographic system is determined by the optical parameters, detector performance, preamplifier's noise, the signal processing system, the picture presentation and evaluation systems.

24.4.1 Sensitivity of Thermographic Imaging System

A figure of merit for the thermographic imaging system is the noise equivalent temperature difference (NETD). This is usually called minimum temperature resolution, i.e., the temperature differential between two adjacent elements in the scene that will give a signal equal to the system noise. Thus, the smaller the NETD, the better the sensitivity.

In order to obtain the minimum value of NETD, the detector with the maximum value of integration should be used. Also, even if the scanning and optical system is the same, thermographic systems with different detectors have different NETD. The peak detectivity of InSb (photo-detector cell) is 1.3×10^{11} and that of CdHgTe, is 2×10^9 . Therefore, InSb detector is one order of magnitude more sensitive than CdHgTe.

24.4.2 Recording Techniques

Several techniques are used for making photographic recording of the image from the screen. Adapters for 35 or 60 mm film and Polaroid cameras are available to suit different needs. For quantitative evaluation, the usual procedure is to take one photo for each isotherm level.

For recording dynamic changes in thermal patterns, a 16 mm cine can be used. Cine pictures are useful when studying changes in peripheral circulation due to exercise and changes in environmental temperature. It is also possible to make a video recording on video camera and play it back to the display unit for later evaluation. For colour recording, each isotherm setting is exposed through a different colour filter on a single frame of colour film. This gives concentrated and easily interpreted information on the temperature distribution. Each colour corresponds to a specific temperature.

■■■■► 24.5 QUANTITATIVE MEDICAL THERMOGRAPHY

For comparing the results of successive thermographic examinations, it is essential that the results are standardized and quantified. In the earlier thermographic equipment, the thermograph was recorded on a photographic film from which it was possible to quantify it by densitometry. Work with this system was limited by the long scanning time. A practical solution to this problem is the use of 'isotherms'. Differences between the various gray tones are determined accurately by means of a thermal band or isotherm. This is visualized by calibrating the electronic circuitry inside the camera, so that a particular gray level (i.e., temperature) is depicted as bright white on the screen.

Thus, the isotherm is built up from white areas each indicating the selected temperature. These isotherms can be shifted to any temperature level and will register temperature differences as small as 0.2°C which are read from a scale on the monitor screen. The use of isotherms thus provides a series of points between selected temperatures providing a thermal contour map. A single isotherm could be displayed as a marker of maximum or minimum temperature, in high contrast from the surrounding temperatures. The calibration of isotherms is possible by scanning a temperature reference block.

In the modern thermographic equipment, temperature measurement is improved by providing two simultaneous isotherm functions. One isotherm could be located on an external temperature reference and the other on a point of interest. The unknown temperature is determined by difference. A convenient method of quantitative recording is the superimposition of a range of isotherms in a 7, 8 or 10 colour-coded sequence between 26 and 34°C , representing the temperature range from the normal to the grossly inflamed, yellow usually represents the maximum temperature and blue the minimum. The changes can then be quantified by an integrated numerical expression of the isothermal images. For advanced medical thermography, the modern systems include facilities for both analog as well as digital analysis for automatically measuring complex temperature levels and for defining temperature areas.

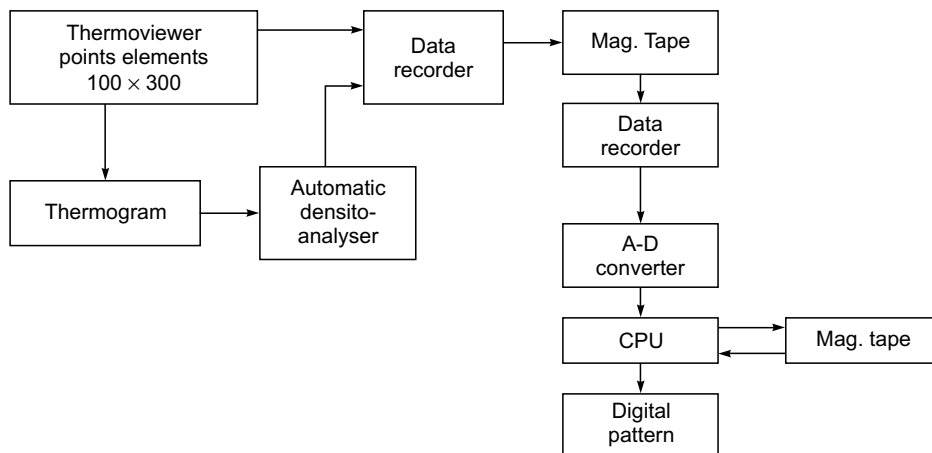
24.5.1 Digital Analysis of Thermograms

For complex image patterns, computers offer excellent opportunities for efficient and objective analysis. Thermal images can be digitally analysed in many ways and for many reasons. Firstly, it can be used to determine numerous parameters from the image itself, highest, lowest and average temperature, or differences between one region and another or areas of various temperature contours or geometric centroids or skewness and so on. Secondly and more importantly, the computer can be programmed to analyse these parameters and plot them as contours or gray tones or isothermal maps or other formats. Computers can also be used to draw conclusions from these parameters, and based on user-defined algorithms solve complex analysis problems. Alternatively, the computer may be used for automatic interpretation of the information received.

There are also more complex image computations that are frequently of interest in thermographic applications. Depending on the speed required and the complexity of the computation, the desired function may be performed by the digital image processor.

Digitizing Thermal Images: A direct LCD display of a thermogram permits only about 8 shades of gray to be distinguished between black and white, and a photographic record from a LCD face provide a permanent record of only 5 or 6 shades. Even a colour pattern has only a 10-step resolution. Thus, when a wide temperature range is measured, as sometimes encountered in clinical situations, it is difficult to distinguish small temperature differences. Besides this, absolute surface temperatures and temperature differences between two points on the body surface cannot be measured directly and precisely in a photographic thermogram. By using computer techniques or a microprocessor in a dedicated system, it is possible to have a very large range of gray levels.

Fig. 24.7 shows a schematic diagram for digitizing a thermogram (Fujimasa *et al.*, 1973). The output signal of the infrared radiation camera is recorded on a magnetic tape. Analog outputs from the camera are converted into digital thermo-profiles by the A-D converter of a computer. The control processing unit of the computer converts digital values into true temperature values based on the calibration data stored in the memory or magnetic tape. Alternatively, the thermographic patterns can be scanned and converted into analog electrical signals by a scanning densitometer and are converted into digital form. For a precise analysis of surface temperatures, the thermogram digitization and computers are useful in the automatic analysis of thermograms.



► Fig. 24.7 Schematic diagram for digitization of a thermogram (Redrawn after Fujimasa *et al.*, 1973)

Computer analysis is provided as a standard option with modern thermographic equipment. This option includes not only hardware but also software. By converting the analog image into a digital form and then storing it on magnetic tape, the patients' thermal image is always available for live-like analysis, irrespective of when it was taken. There is a wide range of software available for statistical data analysis, breast thermal analysis and complex image analysis.

24.6 PYROELECTRIC VIDICON BASED THERMOGRAPHIC CAMERA

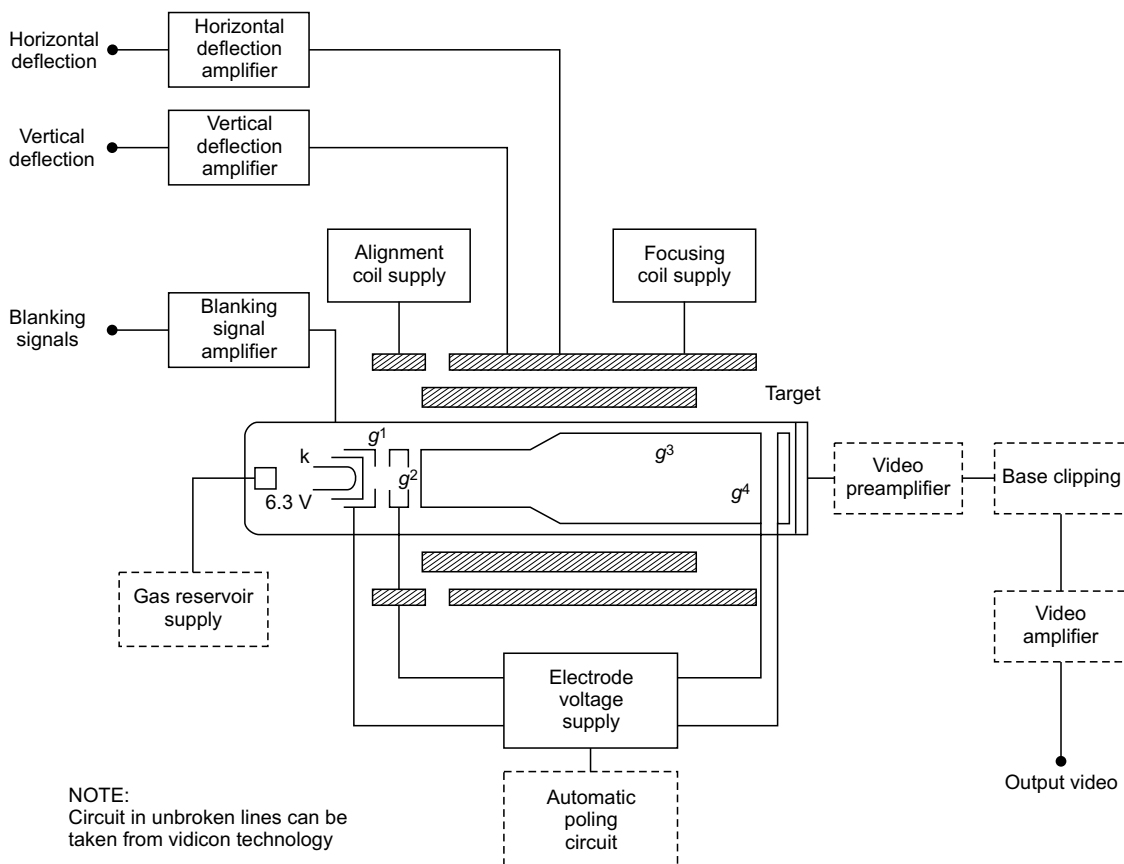
The pyroelectric vidicon is a thermal imaging tube. This tube resembles the standard vidicon, the main difference being that the material used for the target is sensitive to infrared instead of visible light. It is a compact, sensitive and reliable device and can thus find convenient application in medical thermography.

The pyroelectric vidicon basically consists of: (i) a glass envelope; fitted with a germanium faceplate, matching the 8 to 14 μm atmospheric window, (ii) a pyroelectric target mounted on a metal backing plate that also acts as the video output electrode, (iii) an electron gun and beam shaping electrodes like those of a standard vidicon, (iv) a gas reservoir heated by a tungsten filament. The tube is surrounded by focusing and deflection coils that are identical to those used with a standard vidicon.

Incoming thermal radiation is focused by an infrared transmitting lens through the infrared transmitting face plate of the tube onto the target. The target is a triglycine-sulphate (TGS) pyro-electric target which has a high sensitivity in the 8 to 14 μm range of the spectrum. Certain types of crystals (ferro-electric) get polarized in a well-defined direction, known as the polar axis. Because the degree of polarization is temperature-dependent, heating or cooling a slice of such a crystal will create an accumulation of charge (on the faces normal to the polar axis) that is proportional to the variation in polarization caused by the temperature change. This is called the pyroelectric effect, characterized by the pyroelectric coefficient $P = \text{db}/\text{dt}$. The increase in pyroelectric coefficient with temperature is accompanied by a very rapid increase in the dielectric constant of the material that reduces the efficiency with which the pyroelectric changes can be read by the electron beam.

For this reason, the maximum useful pyroelectric effect is obtained with a target temperature of about 35°C. It must never exceed 40°C, the curie point of TGS. Also, the electrical changes are produced only when the temperature of the pyroelectric material changes. So, no image output will be produced from an unchanging image. It is for this reason that the incoming radiation must be chopped or the camera panned so as to produce target temperature variations. The pyroelectric materials also exhibit piezo-electric properties. Therefore, to avoid signal degradation by micro- phonic noise, excessive vibrations should not reach the camera.

Pyricon is the name given to pyroelectric vidicon (PEV) manufactured by Thomson CSF. Fig. 24.8 shows a block diagram of the electronics of a camera using the pyricon. Most of the circuit blocks are similar to the vidicon camera used in closed circuit TV systems. There are, however, two special circuits; the pre-amplifier and video amplifier. The output signal current from the pyricon tube is low (0 to 50 nA). So, a high current gain, low noise pre-amplifier must be used. The amplifier should also have a bandwidth matched to the maximum spatial frequency and to the scanning standard. A typical pre-amplifier would have a bandwidth of 2.5 MHz and a load resistance of 3.3 M Ω . The video amplifier should have an adequate gain to match the low value of the useful signal. It is generally arranged to give a gain of 10 to 60. In a typical case, a temperature change of 1°C gives a tube signal current of around 1 μA . This would give a pre-amplifier output signal of 3.3 mV. If the video amplifier has a voltage gain of 60, its output signal will be $3.3 \times 60 = 200$ mV. A base clipping circuit is provided in the pyricon camera. This is used for block level adjustment, by setting the pedestal level at the proper black level of the picture. A pyricon camera provides 0.5°C resolution at ambient temperature. The common errors in such systems are pedestal variations noise and thermal blurring. The PEV thermograph is silent like a TV camera and has full access to all available TV equipment and accessories.



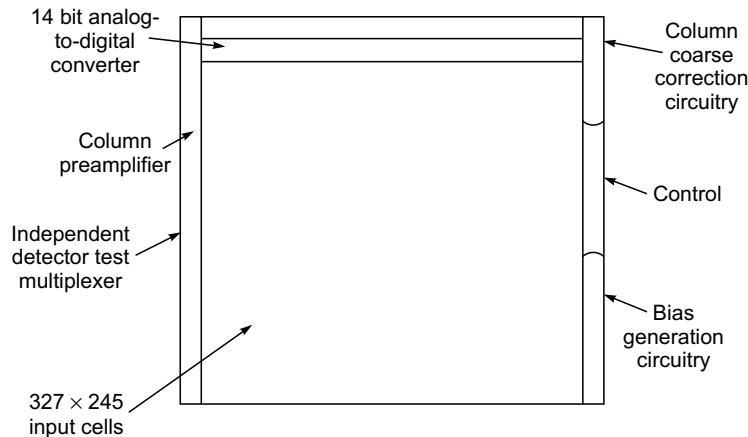
► **Fig. 24.8** Block diagram of camera electronics used with Pyrricon (Courtesy: Thomson CSF, France)

24.7 THERMAL CAMERA-BASED ON IR ARRAY SENSOR

Uncooled infrared technology is revolutionizing IR imaging by offering low-cost sensors with thermal sensitivity comparable to many high performance and complex cryogenically cooled infrared detectors. These sensors make use of micro-bolometer technology which involves a silicon micro-machined sensor that uses wafer-level silicon processing to fabricate a thermal sensor. White *et al.* (1998) give details of such a sensor and that of a thermal camera-based on this technology.

A bolometer is a thermal detector heated by incident radiation, resulting in temperature rise that is sensed as a change in the element resistance. The 327×245 micro-bolometer focal plane array (FPA) has micro-machined bolometer elements on a 46.25 mm pitch. Each micro-bolometer detector consists of a silicon nitride micro-bridge that lies above a CMOS silicon substrate and is supported by two silicon nitride legs. A vanadium oxide film which has an approximately 2% temperature coefficient of resistance at ambient temperatures is deposited on the bridge to form the bolometer resistor. Each of the microbolometer detectors is connected to an underlying unit cell in the silicon CMOS readout integrated circuit substrate via two holes in the passivation

layer on the top of the integrated circuit. A layout of the complete integrated circuit is shown in Fig. 24.9. A unique feature of a bolometer array is pulse biasing, in which the bias necessary to measure the element resistance is applied for a very short time. This allows the basic unit cell to be simple with most of the complex circuitry located in the periphery of the chip.



► **Fig. 24.9** Layout of an integrated circuit used as a micro-bolometer focal plane array (redrawn after White *et al.*, 1998). Width is approximately 17 mm.

The camera based on FPA is a system with functionality allocated between focal plane array and electronics to optimize performance. The analog-to-digital converter and a number of other functions are incorporated on the FPA, while enabling the use of all digital electronics of the FPA. With the fabrication of detectors directly on silicon and application of silicon process techniques at wafer level, millions of mechanical structures can be fabricated on a single silicon wafer at a very low cost.

The imaging performance of this camera shows linearity better than 0.25% over the range from 5°C to 100°C, a thermal constant of 14 msec and temperature coefficient of resistance as 1.5–2%/K. The IR absorption in the 8–12µm range is 80%. Video output is in NTSC or PAL format.

Fig. 24.10 shows a digital infrared thermal imaging system in use. The system has two parts, the IR camera and a standard PC or laptop computer, making the system portable. The thermal sensitivity of the camera manufactured by M/s Meditherm Inc. is less than 0.01°C with an acquisition temperature range of 15°C to 40°C and field of view as 30° × 22°. The detector consists of amorphous Silicone Microbolometer Focal Plane Array (FPA) 160 × 120 with spectral response of 7–14 microns.

A thermal imaging camera consists of five components: an optic system, sensor (detector), amplifier, signal processing electronics, and display. These components work together to render infrared radiation into a visible light representation in real time. The camera displays infrared output differentials, so two objects with the same temperature will appear to be the same “colour”.

The system can measure temperatures ranging from 10°C–55°C to an accuracy of 0.01°C. Focus adjustment covers small areas down to 75 × 75 mm. Modern thermal imaging is now available using small camera systems that can store digital images, or may be used online with a portable or desktop computer. A number of advanced image processing techniques assist



► **Fig. 24.10** Infrared camera in use for digital infrared thermal imaging system (Courtesy: M/s Meditherm Inc.)

the operator in achieving standardized and quantifiable images of temperature distribution. Both knowledge of the practical applications for thermal imaging, and the provision of modern imaging systems have dramatically improved in recent years. Good software with enhancement and analysis is now available in thermal imaging.

Thermograms are stored as TIFF images, thus permitting access and manipulation with a wide range of off-the-shelf application software. In addition, the thermal imaging software not only controls the operation of the IR camera but also provides features including colour scale control, base temperature control, display spot temperature, statistical data for a user defined area, temperature profiles, image magnification and colourmap selection. Because PC based software is used, the system can hold vast amounts of data on thousands of patients.

Unlike most diagnostic modalities, infrared thermographic imaging system is non-invasive and is capable to show physiological change and metabolic processes. It has proved to be very useful complimentary procedure to other imaging and diagnostic modalities.

MODEL QUESTIONS

1. What is the basis of thermal imaging system? Draw a diagram showing spectral distribution of infrared emission from human skin with respect to visible range.
2. What type of detectors are used in thermal imaging system? Illustrate the spectral characteristics of various types of infrared detectors. What are solid state detectors and what are their advantages?
3. Illustrate the construction of a pyroelectric vidicon camera. What is the detector used in these systems?